



Notes

Apr 23, 2026

Meeting Apr 23, 2026 at 21:07 EEST

Meeting records  Transcript

Summary

The team established a weekly study group for Lean 4 formal verification via collaborative code-first sessions.

Study Group Operational Structure

The group adopted a guided session format requiring pre-session preparation to maximize time for complex material. Members will rotate as weekly leads to summarize chapters and review code.

Technical Workflow and Resources

Participants will store meeting notes and code exercises on GitHub while coordinating discussions via Discord. The team confirmed a focus on formal verification and theorem proving.

Tooling and Learning Roadmap

The group decided to utilize the Natural Number Game and Mathlib to progress through Lean 4 concepts. They scheduled weekly two-hour sessions for interactive learning and code review.

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Decisions

We've **updated the Decisions section** using your feedback.

Let us know what you think: [Helpful](#) or [Not Helpful](#)

ALIGNED

- **Discord resources channel creation** A dedicated Discord channel will be established for the team to store and share links and resources.
- **Code-first learning approach** The learning approach will prioritize coding and applying examples over theoretical reading.
- **Just-in-time chapter planning** Chapter pacing and splitting will be determined on a per-session basis rather than mapped out in advance.
- **Guided session presentation structure** Sessions will involve all members reading material in advance, with one designated person presenting a summary and leading the discussion each week.
- **Code demonstration and repository management** The session guide will demonstrate their code during the meeting, and relevant code will be pushed to the main repository for tracking purposes.
- **Session scheduling and planning** A shared document will be used to manage the schedule and assign members to lead specific chapters.
- **Platform tools selected** GitHub is chosen for central file storage, Discord is retained for text chat, and Google Meets is selected for video sessions.
- **Meeting schedule established** Meeting cadence is established as weekly, two-hour sessions occurring on Thursdays.
- **Agenda planning process formalized** Session structure and agenda planning are to be conducted on Discord prior to each meeting to ensure organization.
- **Introduction session added** A five-minute introduction round for participants to share backgrounds and goals is added to the agenda for the next session.
- **Planning deadline established** Planning for upcoming session material and identifying a session guide must be completed by every Friday.

- **Study method set to individual preparation** Reading and independent coding exercises are to be completed individually before the session rather than during the meeting.
- **Primary textbook selected** The book 'Functional Programming in Lean' is confirmed as the primary study material for the group.

Next steps

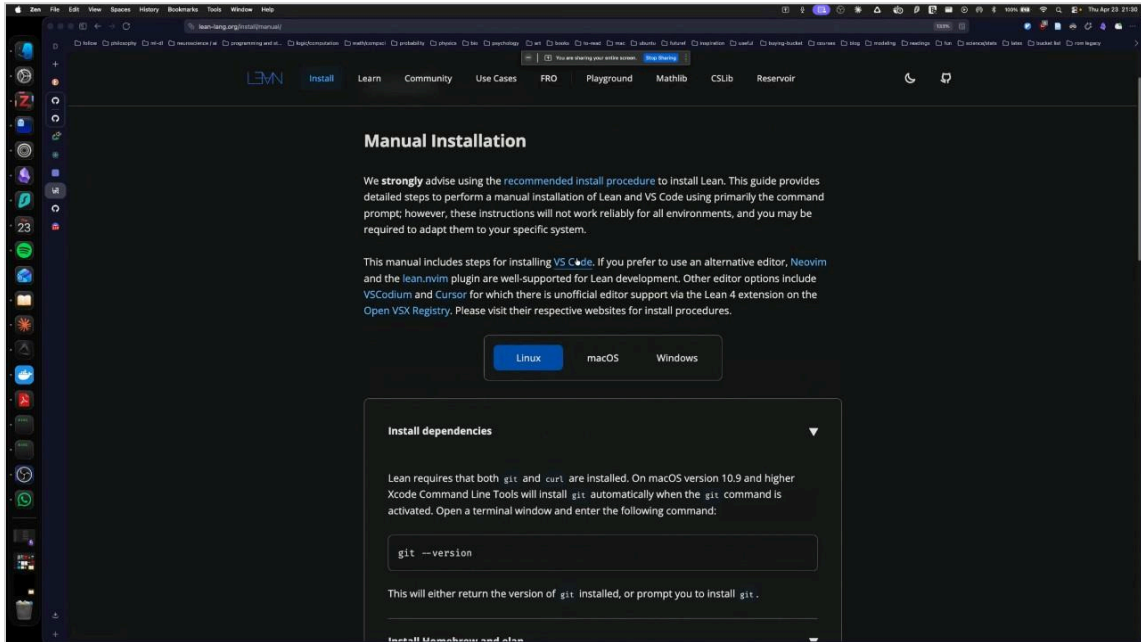
- ☐ [Claudiu Leoveanu-Condrei] Export Notes: Export current meeting notes. Place files onto the GitHub repository.
- ☐ [Claudiu Leoveanu-Condrei] Discord Channel: Create a resources group channel on Discord. Use this for sharing links.
- ☐ [Claudiu Leoveanu-Condrei] Share Resources: Distribute relevant links including installation guides, the MATL package, the natural number game, and the Deep Wiki resource. Upload additional Lean books to GitHub.
- ☐ [Claudiu Leoveanu-Condrei] Check Permissions: Investigate Discord roles and permissions. Determine how to allow other members to create channels.
- ☐ [Luiza Corpaci] Zulip Group: Investigate creating a private group channel on the Zulip Lean community forum.
- ☐ [Claudiu Leoveanu-Condrei] Track Time: Monitor session length; enforce maximum 2-hour limit.
- ☐ [Claudiu Leoveanu-Condrei] Send Invites: Distribute calendar invites for weekly sessions.
- ☐ [The group] Define Agenda: Decide material scope and assign session guide by Friday.
- ☐ [Luiza Corpaci] Update Calendar: Add agenda structure details to meeting invite description.
- ☐ [Claudiu Leoveanu-Condrei] Distribute Resources: Share correct Lean 4 book link and Matlib resource.
- ☐ [Claudiu Leoveanu-Condrei] Locate Protocol: Find and send standardized business logic mapping format.
- ☐ [Claudiu Leoveanu-Condrei] Upload Resources: Drop all links and meeting notes into the GitHub repository.

Details

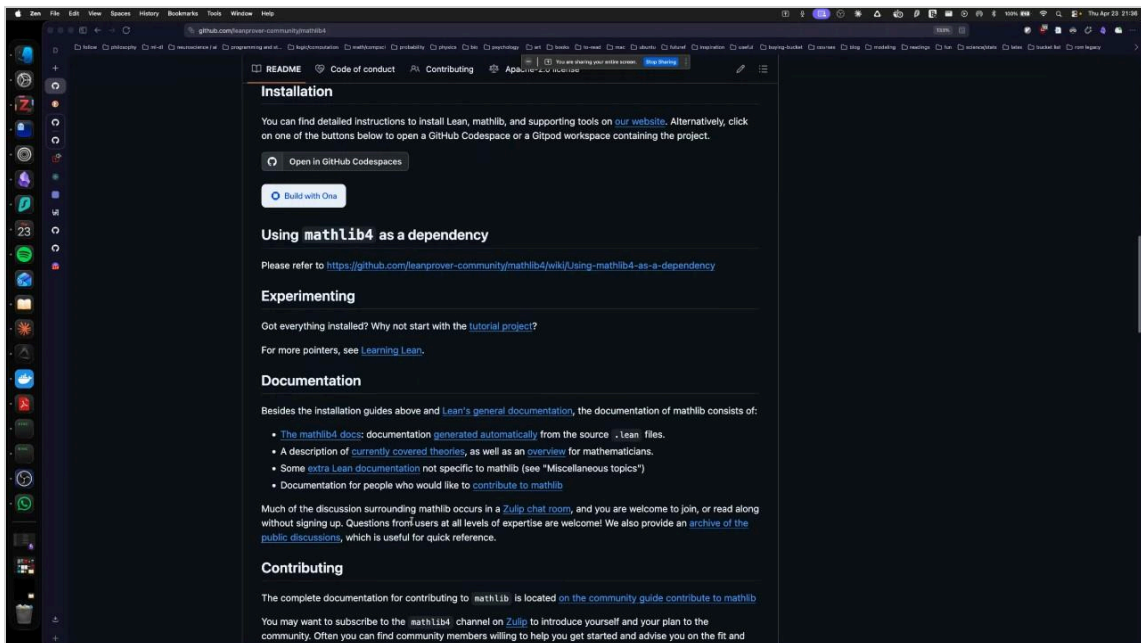
Did the screenshots in this section make your notes [better](#) or [worse](#)?

- **Discussion on Meeting Notes and Resources:** Claudiu Leoveanu-Condrei plans to set up a folder on Google Drive for sharing meeting notes, though they are unsure if the current setup will automatically distribute them via email. Ultimately, they decided that notes and resources will be stored on GitHub, with links shared on Discord in a dedicated "resources" group channel ([00:00:00](#)). They also noted that Nelson and Adrien, two individuals working with them, will join the group starting next week, though Adrien's arrival might be delayed due to travel ([00:01:08](#)).
- **Structure of the GitHub Repository:** The group decided to base their GitHub repository structure on an existing, successful model used by other developers, specifically citing the work of Jeremy Howard. Claudiu Leoveanu-Condrei intends for the repository to mimic this structure, which looked cool and was well-organized ([00:02:18](#)) ([00:04:17](#)). Robert Maria requested that the screen share be zoomed in a bit for better visibility on a laptop ([00:03:22](#)).
- **Long-Term Group Vision and Guest Lecturers:** The group aims to progress through the material to the point where they can invite experienced individuals who actively use Lean for topics like physics or algorithms to give presentations ([00:04:17](#)). Claudiu Leoveanu-Condrei emphasized that they are content with a small group (four to six people) actively progressing through the basics ([00:05:25](#)). They assume people are more likely to commit to the group after seeing some initial lectures or material ([00:04:17](#)).
- **Preference for a Code-First Learning Approach:** Claudiu Leoveanu-Condrei proposed a learning approach that focuses on coding examples alongside the book, noting that as a developer, they find reading code beneficial for understanding complex concepts ([00:06:33](#)). Andrei Ungureanu supported this, stating that going through code is the best way to learn, and Luiza Corpaci confirmed that the goal is to map the book's content into code exercises and save them to GitHub. Robert Maria mentioned having previously coded examples from the book but feeling like they missed the context of Lean's purpose ([00:07:49](#)).

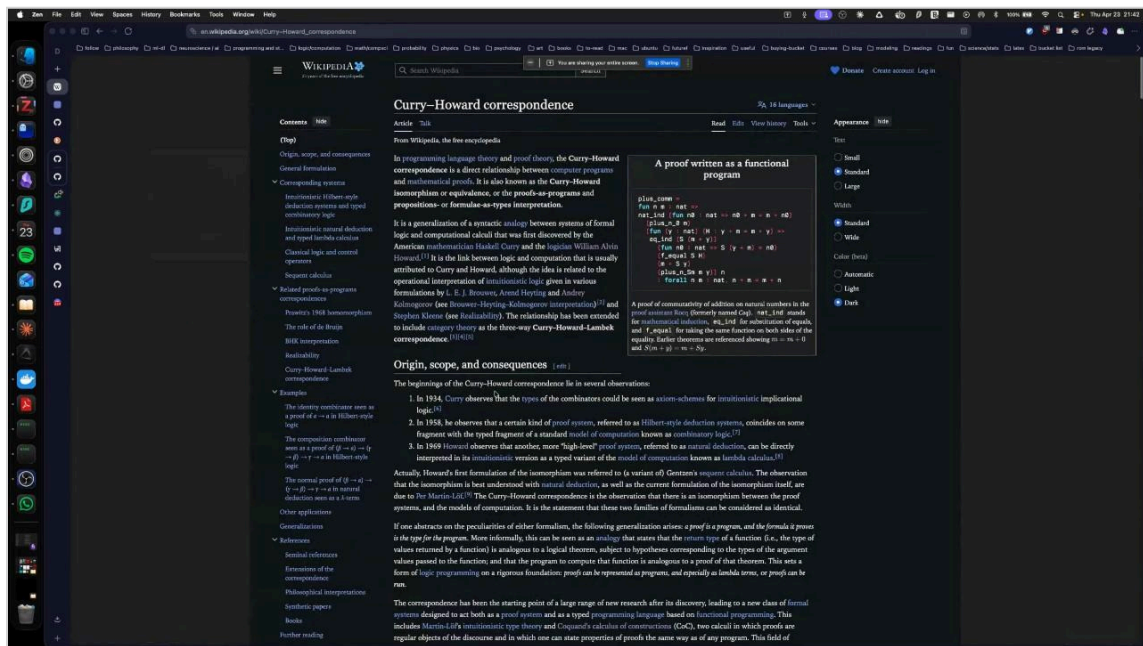
- **Structuring the Book Coverage:** The group agreed to follow the well-structured nature of the book, but acknowledged that chapter lengths will vary, requiring some to be split across multiple sessions ([00:08:37](#)). They decided that they should prepare in advance, mapping out the necessary chapter splits beforehand rather than mapping the entire book in advance. This approach will allow them to focus more on one concept when coding examples ([00:09:46](#)).
- **Discussion on Session Format and Book Reading:** Andrei Ungureanu suggested that they prepare in advance for sessions and focus meeting time on complicated material or concepts that were not understood, rather than reading the book aloud ([00:10:45](#)). Claudiu Leoveanu-Condrei agreed with this proposed approach, seeking input from Luiza Corpaci and Robert Maria ([00:11:59](#)).
- **Adoption of a Guided Session Format:** Luiza Corpaci suggested a format where everyone reads the chapter beforehand, and one person prepares a short summary or highlights open issues and complex points for discussion. This "guided session" approach was accepted by Claudiu Leoveanu-Condrei, and the group agreed to alternate who leads the discussion each week ([00:11:59](#)).
- **Robert Maria's Departure and Teaching Philosophy:** Robert Maria, who described himself as having a teaching background, had to leave the meeting early due to a personal matter ([00:39:04](#)). Before leaving, they emphasized that students learn best by doing, using examples, and with live coding demonstrations ([00:14:58](#)).
- **Discord Channel Organization and Permissions:** Luiza Corpaci clarified that the "resources" channel on Discord should be used as a link dump and not for general discussion, suggesting a separate discussion channel ([00:15:57](#)). Claudiu Leoveanu-Condrei plans to look into setting up roles and permissions to allow Luiza Corpaci to create or manage channels, as they could not do so initially ([00:17:04](#)).
- **Installation and Tool Recommendations for Lean:** Claudiu Leoveanu-Condrei recommended that everyone complete the installation of the necessary tools, noting that VS Code is the most robust option, with NeoVim also supported for a good experience ([00:18:11](#)). They also mentioned that Cursor, a VS Code fork, is compatible. Andrei Ungureanu and Luiza Corpaci confirmed they had already installed the necessary tools ([00:20:25](#)).



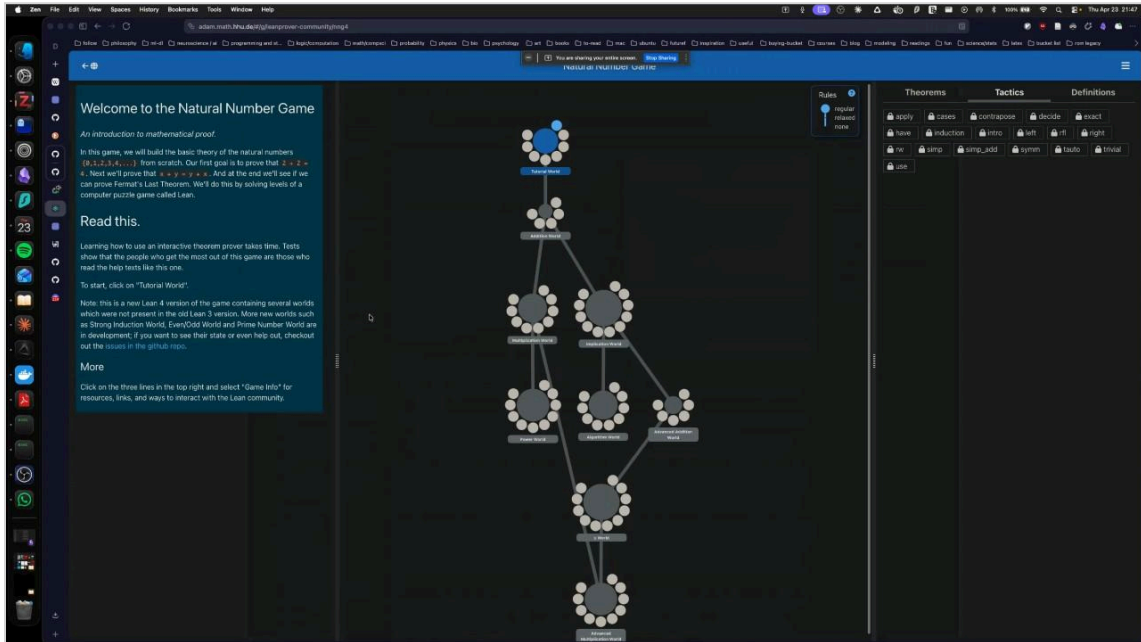
- **Installation of the Mathlib Package:** The discussion shifted to the necessity of installing Mathlib, a constantly advancing community package that formalizes mathematics and is used by many libraries. The group was advised to install the Lean version manager (lean-toolchain) and Mathlib, noting that the latter is built using `lake` and acts as a single, core dependency for proofs ([00:22:36](#)).



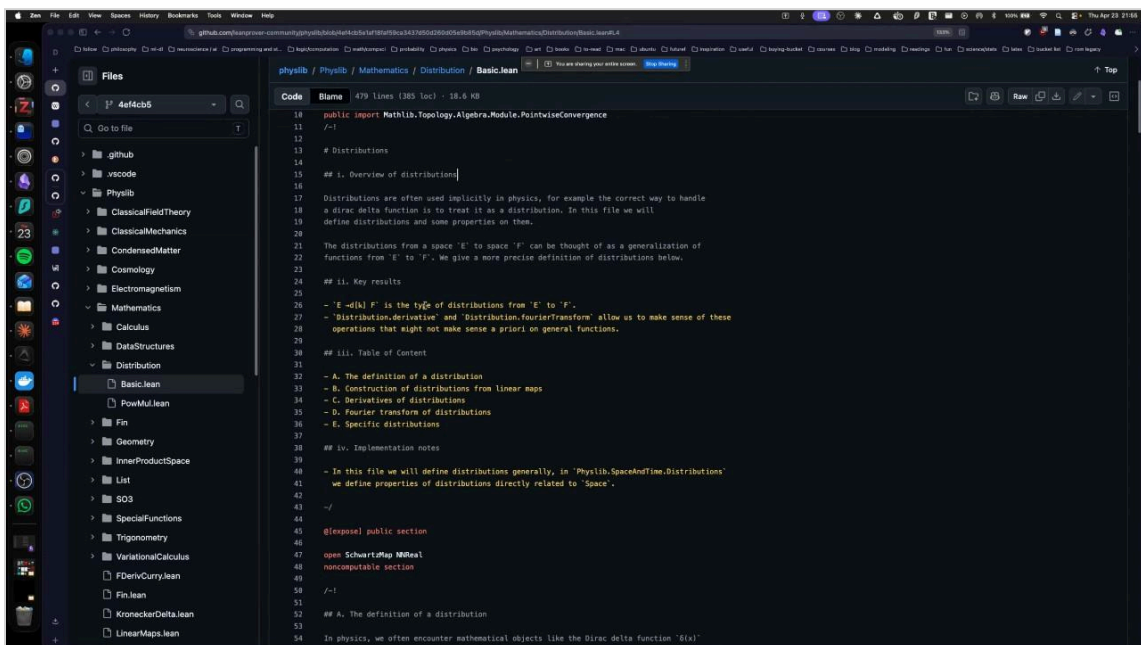
- **Understanding the Uniqueness and Power of Lean:** Robert Maria inquired about the distinct features of Lean compared to other functional programming languages and theorem provers ([00:27:12](#)). Claudiu Leoveanu-Condrei explained that Lean 4 is both a Turing complete programming language and an automated theorem prover, differentiating it from SMT solvers like Z3 ([00:28:31](#)). The power of Lean is fundamentally tied to the Curry-Howard correspondence, which posits that a proof is a program ([00:30:10](#)).



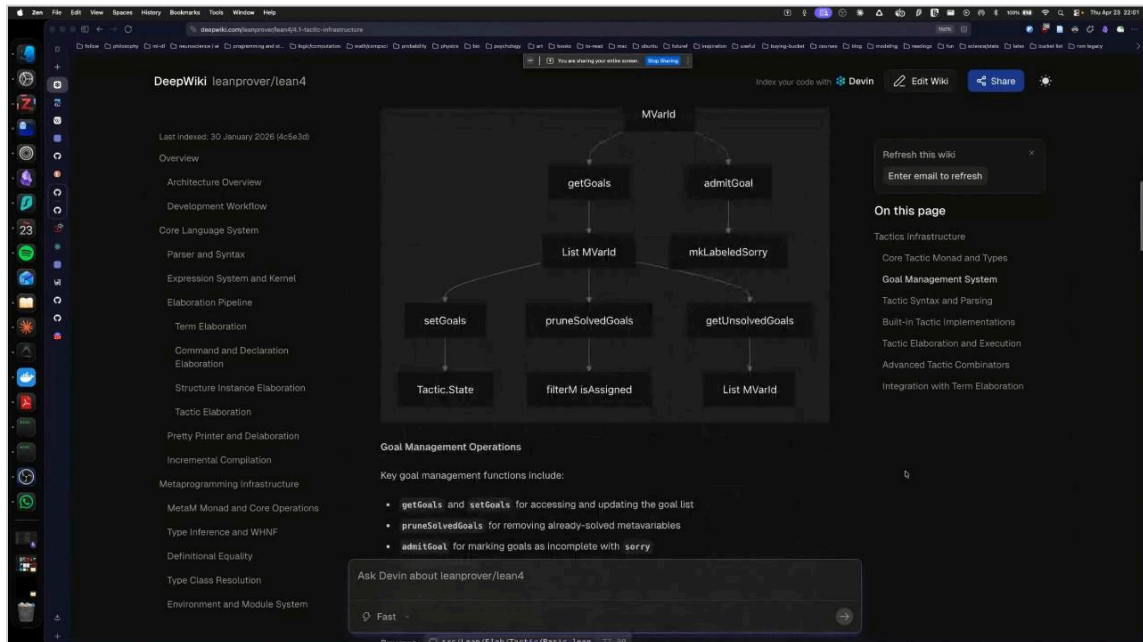
- **Dependent Type Theory and the Natural Number Game:** Claudiu Leoveanu-Condrei explained that Lean 4 is a dependent type language, requiring every variable and statement to be explicitly typed, making the typing integral to the argument structure ([00:31:46](#)) ([00:34:38](#)). They recommended the "Natural Number Game," an interactive tutorial that teaches the fundamental Peano axioms and the use of "tactics" to prove statements, noting that proving tautologies requires explicit steps ([00:36:03](#)).



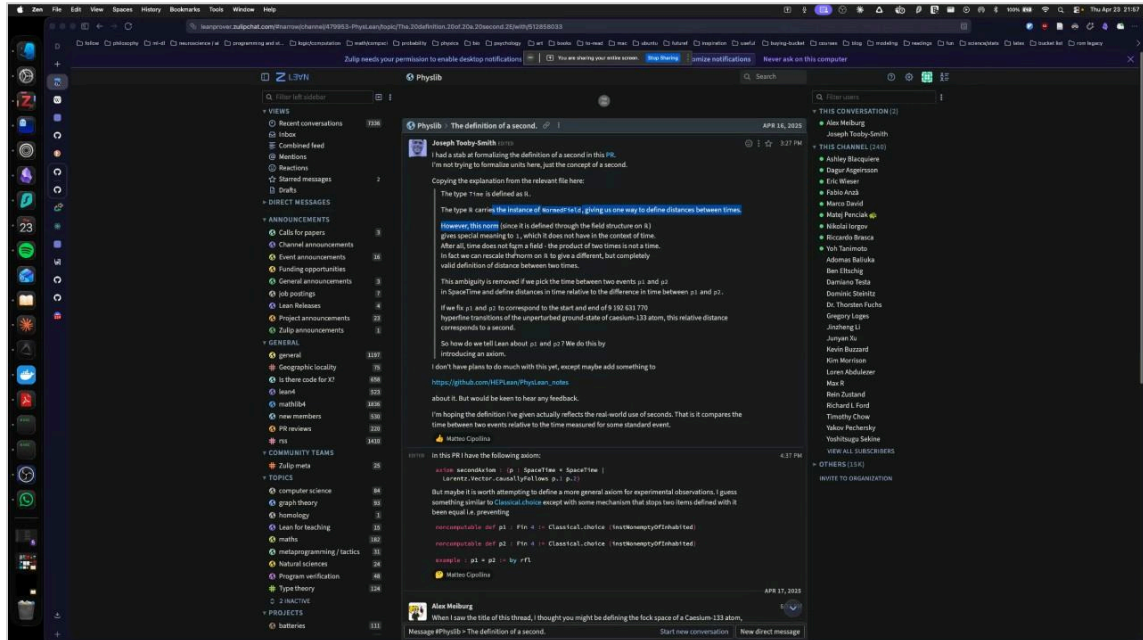
- Discussion on Formalizing Physics in Lean:** Claudiu Leoveanu-Condrei shared their interest in projects formalizing physics in Lean, citing an example where they were reviewing the formalization of distributions ([00:41:52](#)). They also highlighted the extensive, explanatory comments in the Lean main library and projects like `Lean and Mathlib Physics`, noting the complexity of formally defining basic concepts like a "second" ([00:43:30](#)).



- Recommendation for Deep Wiki as a Resource:** Claudiu Leoveanu-Condrei highly recommended using the Deep Wiki tool, built by the Cognition team, as a great resource for parsing complicated projects and navigating Lean 4 code (00:46:58). They mentioned that Deep Wiki provides diagrams and an organized view of the code, making it particularly useful for those who learn better visually (00:48:24).

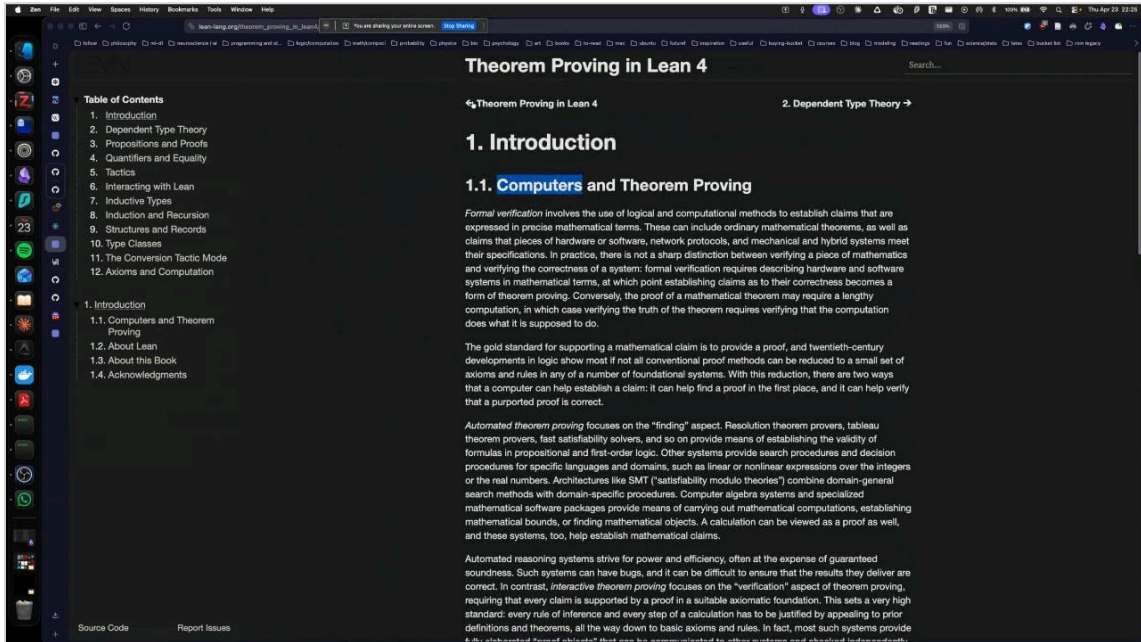


- Approach to Code Writing and Tracking:** Luiza Corpaci asked how the code writing should be approached, leading to an agreement that the group should write the code for the exercises beforehand, and then discuss it during the session (00:50:58). Regarding GitHub tracking, Andrei Ungureanu suggested that the guide for the session should push their code to the main repository, while others can keep their own repositories, a simple process Claudiu Leoveanu-Condrei supported (00:52:05).
- Engagement with the Official Lean Community (Zulip):** Claudiu Leoveanu-Condrei inquired about joining or creating a channel for the group on the official Lean community platform, Zulip, citing it as a resource for professional expertise and recruitment (00:55:23) (00:57:08). Luiza Corpaci offered to look into how to set up a group on Zulip, as they are already a member there (00:56:19) (00:58:05).

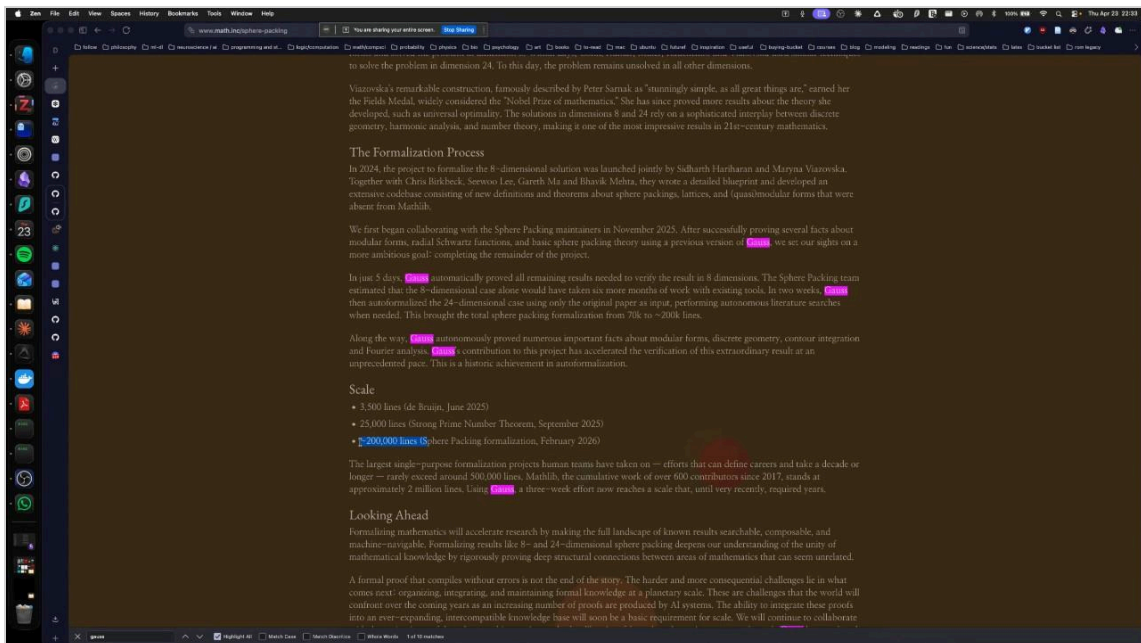


- Scheduling the Session Guide Role:** Luiza Corpaci asked if the assignment of the session guide role should be done week-by-week or planned in advance across the chapters. The group concluded that planning the guide for the chapters in advance would be acceptable (00:58:05).
- Storing Meeting Materials:** The participants decided that the best location for meeting materials, such as tables, is the `readme` file on GitHub, as Claudiu Leoveanu-Condrei prefers to keep everything on GitHub for ease of access. Discord will be used for chat, while the regular meetings will be used for sessions (00:59:07).
- Meeting Frequency and Duration:** It was determined that the reading and discussion sessions will be held weekly to maintain momentum, with the next session scheduled for the following Thursday at the same time. The goal is to keep sessions within a two-hour time frame, such as from 9 to 11, and Claudiu Leoveanu-Condrei offered to monitor the time to prevent derailment (01:00:19).
- Developing a Session Structure:** Luiza Corpaci suggested creating a draft agenda or minimal structure for the two-hour sessions to help manage time effectively, possibly allocating time for general discussion, code review, and open topics. Claudiu Leoveanu-Condrei agreed that having a structure beforehand is necessary and proposed discussing and finalizing the agenda on Discord before the session (01:01:18).

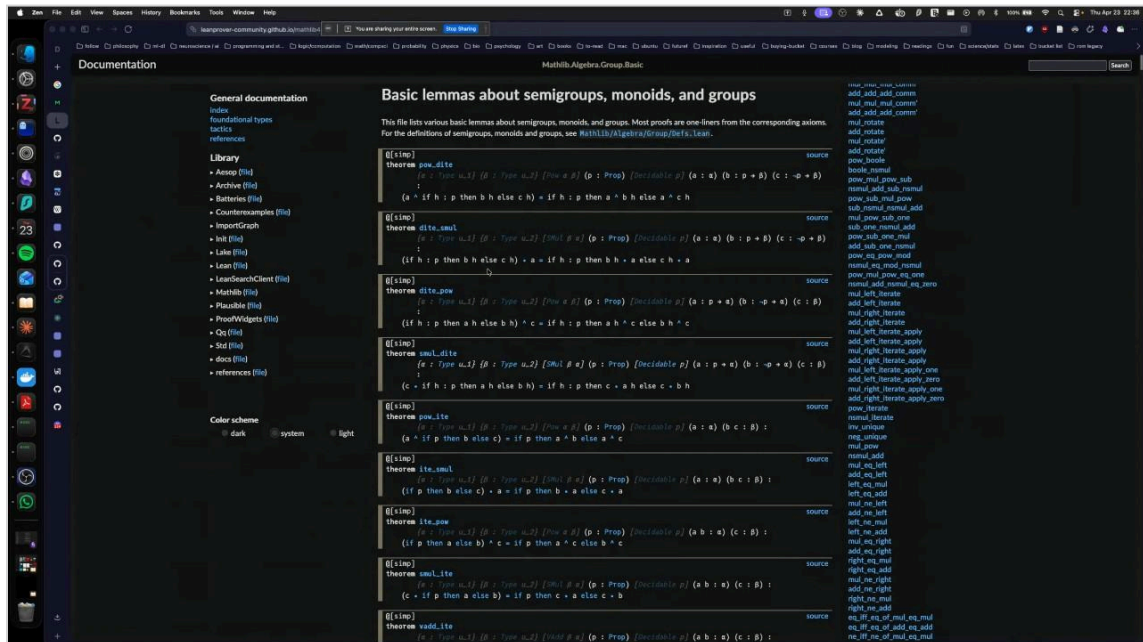
- **Upcoming Participant and Introductions:** Claudiu Leoveanu-Condrei noted that Nelson, who has a PhD in symbolic logic, will be joining the sessions, and Luiza Corpaci suggested adding a short "mini spot" at the beginning of the next session for participants to describe their motivation and goals for using Lean ([01:02:15](#)). They agreed to dedicate about five minutes at the beginning of the next session for these introductions ([01:03:33](#)).
- **Deciding Next Week's Material and Guide:** Andrei Ungureanu brought up the need to decide on the material from the book for the next week, which Claudiu Leoveanu-Condrei confirmed should be decided on Discord ([01:03:33](#)). They decided that the material and the guide for the next session must be finalized by Friday so everyone knows what is happening up until the following Thursday ([01:04:48](#)).
- **Session Preparation and Structure:** They reinforced the decision that reading and preparation should be done before the session, as the two-hour meeting time is too short to cover the reading material. The guide for the session will lead the discussion, potentially going through code examples and referencing the book as needed ([01:06:59](#)).
- **Selecting the Starting Material for Reading:** Luiza Corpaci and Claudiu Leoveanu-Condrei clarified a misunderstanding regarding which chapter to begin with, confirming they plan to start with the "Getting to Know Lean" chapter and then Chapter 2, "Her World," and deciding how to split Chapter 2 due to its length ([01:10:35](#)). Claudiu Leoveanu-Condrei mentioned that the entire introductory chapter, which discusses Lean's motivation, is a deep but optional "rabbit hole" for individual reading ([01:12:42](#)).
- **Correcting the Book Reference:** Luiza Corpaci and Andrei Ungureanu pointed out that the wrong book link was shared on Discord, leading to a clarification about the two main uses and corresponding books for Lean 4 ([01:12:42](#)). Claudiu Leoveanu-Condrei explained the books cover the language aspect (functional programming) and the theorem-proving aspect (formal verification), confirming the group's focus is on the latter ([01:13:37](#)).



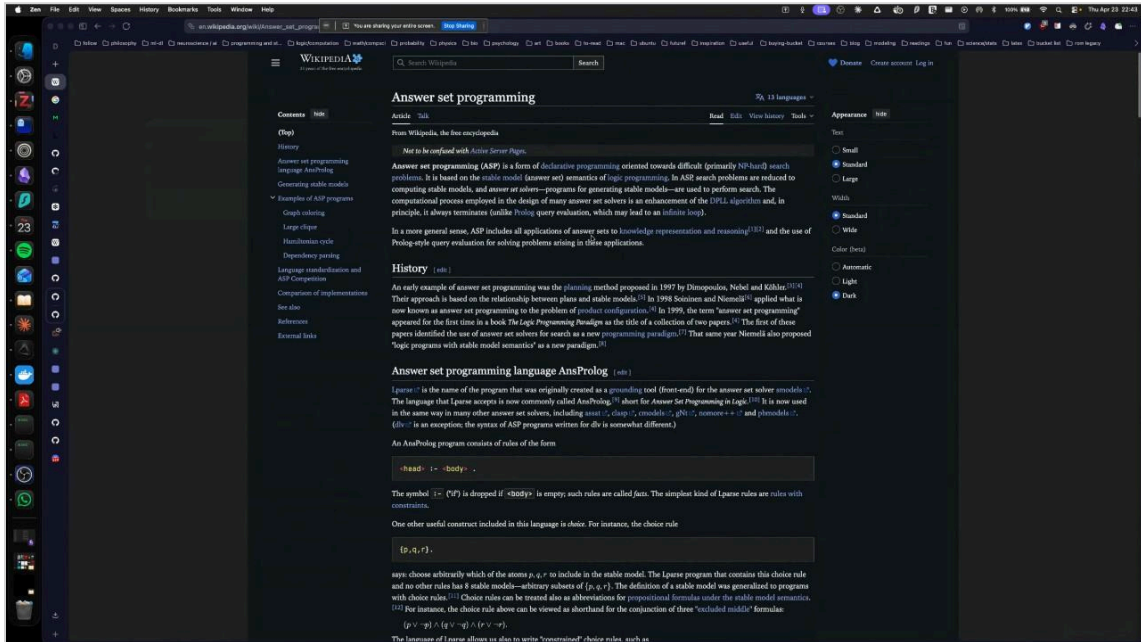
- Scope and Applications of Formalization:** Luiza Corpaci inquired about the limitations and the most complex examples of formalization, leading to a discussion about sphere packing, a formalized field medal result that required 200,000 lines of Lean code ([01:19:06](#)). Claudiu Leoveanu-Condrei also noted that this complexity demonstrates the need for auto-formalization agents ([01:21:48](#)).



- **Resources and Existing Formalized Knowledge:** Claudiu Leoveanu-Condrei shared Matlib as a resource, which is a library containing formalized mathematics that people contribute to and that can be used in proofs, showing that problems in areas like group theory and topology have been formalized ([01:22:56](#)). They noted that people have even used Lean to formalize neural networks ([01:26:09](#)).



- **Personal Motivation for Learning Lean:** Luiza Corpaci stated their interest is in combining formal verification and mechanistic interpretability, while Claudiu Leoveanu-Condrei's motivation stems from security, specifically verifying the output of large language models and formalizing specific protocols ([01:26:09](#)).
- **Discussion on Semantic Diff Checkers:** Luiza Corpaci raised the challenge of creating a semantic diff checker, and Claudiu Leoveanu-Condrei mentioned they have previously implemented this by using logic, specifically answer set programming, as an intermediary compression scheme for natural language text ([01:29:35](#)). Luiza Corpaci is also working on a semantic diff checker using an intermediary representation, aiming to ensure the original intent from a specification file survives mapping into code and test plans, connecting this to the difference between verification (building something right) and validation (building the right thing) ([01:32:02](#)).



- **Concluding the Current Session:** The group confirmed that all links and information are in order, agreeing to close the current session. Claudiu Leoveanu-Condrei committed to sending out weekly invites and beginning to drop the links and notes into the GitHub repository ([01:36:10](#)).

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Transcript

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00:00:00

Claudiu Leoveanu-Condrei: Okay. So, uh I will send you so I will try to make a folder on uh the Google drives so that I can share the notes with all of you. You probably won't get these notes right now. I don't think this is going to work uh because uh you have to be invi I think that's how it works. you have to be invited into a calendar meeting and then you get the meeting notes on the email uh that you join with. But I'm not sure uh it it might work. it you you will all get the the notes for this call, but I'm not sure if it

Luiza Corpaci: Is this your is the meeting like link is that your like personal meeting link

Claudiu Leoveanu-Condrei: will

Luiza Corpaci: that's always the same?

Claudiu Leoveanu-Condrei: uh it's the instant uh one.

Luiza Corpaci: Um,

Claudiu Leoveanu-Condrei: So I guess not. But anyways, don't worry.

Luiza Corpaci: okay.

Claudiu Leoveanu-Condrei: I I I will make anyways a folders. Actually, I probably not even make the folder. we will have everything on the uh GitHub.

00:01:08

Claudiu Leoveanu-Condrei: So I will export the notes so that we have them there as well. Okay. So uh as far as I know we are four and starting next week there will also be Nelson uh and there is also Adrien who's interested and they are both working with me. Um so Adrien is right now in a trip so I'm not sure he will be able to join soon. uh but ness and he will be available starting from the next week. Okay. So let's start with the basics um which any everyone should do in my opinion. Let me quickly share my screen. Actually, before that, let me find the repo that I made so that I can Is it fine if I drop it here or should I put the links on Discord or posterity or whatever you want to call

Luiza Corpaci: Um maybe in the discord and we can have like a group that's called

Claudiu Leoveanu-Condrei: it?

Luiza Corpaci: resources or something like that and then that can be just sharing

Claudiu Leoveanu-Condrei: Yeah,

Luiza Corpaci: links.

00:02:18

Claudiu Leoveanu-Condrei: perfect. We'll do it that way. Okay. So, um where is the share screen? So, there is this um I made this repo. So,

Luiza Corpaci: I cannot get it.

Claudiu Leoveanu-Condrei: Oh,

Luiza Corpaci: Oh,

Claudiu Leoveanu-Condrei: it's loading.

Luiza Corpaci: no. Now it's

Claudiu Leoveanu-Condrei: Okay. So, we will put everything here.

Luiza Corpaci: Yep.

Claudiu Leoveanu-Condrei: Um, it's it's going to look more or less like um like this. So, these are the guys that I copied it from. More or less the name

Luiza Corpaci: Uh, small pause. Currently, it's just the the Oh,

Claudiu Leoveanu-Condrei: um

Luiza Corpaci: okay. Maybe I'm having an issue because I was seeing the Google Meets page until Do you also uh see it with delay? Uh, Robert and Andre.

robert maria: No, for me it's it's fine.

Andrei Ungureanu: It's fine for me. Yes.

Luiza Corpaci: Ah, okay

robert maria: It's just on the laptop if you can just zoom a little bit.

Luiza Corpaci: then.

robert maria: Hello.

Claudiu Leoveanu-Condrei: on the

00:03:22

robert maria: No, no,

Claudiu Leoveanu-Condrei: left.

robert maria: if you can zoom a little bit because I'm on the laptop so I don't see very big

little bit more.

Claudiu Leoveanu-Condrei: Oh, is it fine like this?

Luiza Corpaci: Yes.

robert maria: Just one more control plus please.

Luiza Corpaci: Oh,

robert maria: It will be

Claudiu Leoveanu-Condrei: Where are your glasses, man? Getting old.

Luiza Corpaci: okay.

Claudiu Leoveanu-Condrei: What's happening, man? When should I check? Now I have to sit in the back of the room because for me it's too

robert maria: I Okay,

Claudiu Leoveanu-Condrei: big.

robert maria: next time I will use a projector to to

Claudiu Leoveanu-Condrei: Uh okay, cool. So the the idea here is to basically mimic this uh this guy's structure. I I've been here. I know I know some of the guys. Uh you Robert probably know this guy, right? This is this is our our buddy uh Jeremy Howard. Uh but I know this guy.

robert maria: Yes.

Claudiu Leoveanu-Condrei: I used to work with him. He's super cool.

00:04:17

Claudiu Leoveanu-Condrei: So he made this group uh with yet another guy. I think it was this guy. Uh yeah, these are from US. I don't know them. Uh but I like what they did. Uh it it seemed cool. It looked cool to me. Uh the way they organized it. So I think we should follow the same structure. Uh when they started there was also not a lot of people uh but once they um started having people that used to work with CUDA like a lot so they had developers or um I don't know people from the industry coming giving up lectures uh for half an hour or 15 minutes or something they just showed an algorithm and then they had discussions probably we can do that as well at some point um I guess people are, you know, they, that's my assumption. I think most of the time people just want to see something first before they commit to, you know, to a group they know nothing about. So, uh, that's usually from my experience how people see these things.

00:05:25

Claudiu Leoveanu-Condrei: So I assume that once we have some some lectures here or like some you know we go through some of the material I I would assume that at some point we can we can ask some people that are actually doing lean uh properly to come and um you know have some um some presentation for us. I I've already targeted some people like that they're doing physics in lean there some are doing I don't know algorithmic in lean and it there are a lot of people uh so I I I will try to aim for that as well as we progress uh but I think we should get the basic first um first couple of lectures so that we kind of at least get the idea um how we should think about lean as a whole Uh but I'm personally happy. I I I I don't necessarily want as a a big group. As long as Yeah. We four, five, four, six, we are grinding through this. I'm quite happy. I like it. I like it that I'm not alone.

00:06:33

Claudiu Leoveanu-Condrei: Put it like that. Okay. So, we will have a similar um structure. Um they have a lot of um I would say codes. They focused on code and I was I wanted to ask you guys um whether or not you would also like it that way. I mean um I don't know which one of you are programmers other than Robert. Um but I am a developer so I I'm pretty much into reading code. So to me if I don't understand anything and there is a way to code it then that would obviously be better for my understanding. So I would I would assume is the same for you as well. Um but correct me if I'm wrong because we can have this kind of approach. So as as we progress to the book and the book is quite nice because it has a lot of examples and they show you very specifically what each kind of like um they will go to the syntax and show how to use it. So we can immediately apply it and see go to examples and see exactly how they work and stuff like this.

00:07:49

Claudiu Leoveanu-Condrei: So I would like more this approach if that's fine with you. I don't know if you have any thoughts

Andrei Ungureanu: It's for me it's fun.

Claudiu Leoveanu-Condrei: anybody.

Andrei Ungureanu: I think uh going through code is like the best way to to learn. So

that's the way to do it.

Claudiu Leoveanu-Condrei: Yeah.

robert maria: I didn't read from the book. I just uh coded some examples.

Luiza Corpaci: Yeah.

robert maria: So I don't know what is used for but I I

Claudiu Leoveanu-Condrei: Okay.

robert maria: followed some coding examples but I missed the point what we can do with the tool with

Claudiu Leoveanu-Condrei: Okay. Cool.

robert maria: Lynn.

Claudiu Leoveanu-Condrei: Uh so you you you said what?

Luiza Corpaci: So So

Claudiu Leoveanu-Condrei: Sorry. What is the point?

robert maria: No, I said I I I took the book and I took examples to code to see how it is.

Claudiu Leoveanu-Condrei: Huh?

robert maria: But I I didn't really to know what is used for and why why it's important to

Claudiu Leoveanu-Condrei: Ah okay. Ah we can discuss a bit today about that.

00:08:37

robert maria: accept

Claudiu Leoveanu-Condrei: No worries. So you wanted to say something?

Andrei Ungureanu: I think Raza wanted to say something.

Claudiu Leoveanu-Condrei: Yeah. Exact.

Andrei Ungureanu: No.

Luiza Corpaci: Yeah. So,

Claudiu Leoveanu-Condrei: Yeah.

Luiza Corpaci: uh I I wanted just to make sure that I understood that. So, the goal would be basically to sort of like map the book into into code.

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: exercises and like other things that we think about and then save the um the files

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: in our GitHub.

Claudiu Leoveanu-Condrei: Uh precisely.

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: Yes.

Luiza Corpaci: Yeah, that's work with

Andrei Ungureanu: And and the idea is to uh sorry to interrupt uh is to find

Luiza Corpaci: me.

Andrei Ungureanu: uh like a common path, common goal based on uh what we learned from the book. Um,

Claudiu Leoveanu-Condrei: Uh so from my understanding of the book um it's pretty well structured so I I would just follow it. Um I my question about the example was more that we should definitely do examples as much as possible per each session.

00:09:46

Claudiu Leoveanu-Condrei: Um but I'm not yet sure how we should handle the chapters because obviously some chapters are longer than the others. So some some chapter might take one session and other chapter might got to be split in two. Uh we can do this before. So we can always be prepared in advance for the next chapter. So like uh we already know the next chapter is going to be okay this is kind of too long. Let's split it in two or three. Um you know what I mean? This is what also what this guy did. Uh so we focus more on one concept from the the chapter when we code things. Uh I think that would be as we go on. Um but that that's in my opinion the structure of how we should approach it. But we shouldn't map it in advance. I mean we shouldn't map the book like we do this this this this like that. Um we should just go to the chapters um map the chapter that we're trying to build beforehand.

00:10:45

Claudiu Leoveanu-Condrei: Uh see exactly how many splits we have. Maybe we can do it in one go. um and keep keep doing this and because I assume as we get more knowledge and experience then it's going to be easy to go through some of the things. Uh but what I liked about the book is that it's very it's it's written um it explains very well why you know some things are the way they are and uh this I don't know how we should handle. So should we also go through the book and uh I mean in the session should we go through the book and read from it or we should always keep it as a reference for the session so that we reference it as we discuss through examples. I don't know how you guys see this

Andrei Ungureanu: uh well if we can follow it and maybe prepare bit in advance I think it

would be wiser to go through the stuff uh that are more complicated or that we didn't understand. Um rather than just like read it out loud uh at each session.

00:11:59

Claudiu Leoveanu-Condrei: I agree. Yeah, for me this was also I I agree with this. I don't know what you others think about it. Louisa Roberts

Luiza Corpaci: Uh so my I like the idea. I was at some point uh in um reading club that was on the Google groups uh and like everyone every week so like a specific person every week would basically send a short summary of this chapter with the intention of uh picking the open issues or open problems or like things that were not fully understood or not covered based on their own experience because that was also a group that had many people coming from different sorts of backgrounds and that felt like an interesting concept. So basically just like out of our small group we just like whenever we meet we decide that all of us read the chapter but then one of us gives like a small discussion or like brings up some discussion points about some of the things they um encountered as more complex and then have that as a

Claudiu Leoveanu-Condrei: I'm I'm I'm also fine with this.

00:13:04

Luiza Corpaci: starting

Claudiu Leoveanu-Condrei: Yeah, I I like it. It's like a guides more like guided session.

Luiza Corpaci: Yeah, to some degree.

Claudiu Leoveanu-Condrei: Yeah.

Luiza Corpaci: And then we can like alternate in like every week have someone else and then also spread out the

Claudiu Leoveanu-Condrei: I I like it for me. There's no there's no issue with doing this.

Andrei Ungureanu: Oh yeah. So basically it's a combination of Louisa and my dear right.

Luiza Corpaci: Yep.

Claudiu Leoveanu-Condrei: Yeah.

Andrei Ungureanu: So we each read before but one of us also presents like a short uh summary of the topic.

Claudiu Leoveanu-Condrei: Yeah, I mean yeah I I mean usually when I start reading this stuff I'm I like I like this stuff. So for instance if you take this stuff and you know you're

like just what the hell is calculus of construction? So I just go here and I and I like and I read this and then I go like okay so okay so this is tied to this guy. So who is this guy?

Andrei Ungureanu: You go in the rabbit

00:14:00

Claudiu Leoveanu-Condrei: So this guy did the cock ex I like that a lot.

Andrei Ungureanu: hole.

Claudiu Leoveanu-Condrei: So I I do this a lot and I and I learn a lot doing this stuff you know and probably I mean we got we are here so I assume probably share some of these things too. So uh usually when I start reading and again the book is very I I read some of it and but I I jumped around around it. It's really well written. I mean I I I read a lot of books in my life and this is this is very well written. these guys know their stuff. That's why I even liked about it that we have a book to follow which is not you know that complicated and and so on. And I also found some other books uh which have more examples uh which is good because we can we can do code more from them and I will also share them uh in the in the GitHub at the end of this. So yeah, I don't know.

00:14:58

Claudiu Leoveanu-Condrei: Robert, do you have any thoughts? Because you are the most experienced from all of us. So he he's a

robert maria: Well, yeah. Yeah.

Claudiu Leoveanu-Condrei: teacher.

robert maria: But uh I uh I don't know how students they are working and I don't know how what's the best way to to to learn uh in a team you know I I

Claudiu Leoveanu-Condrei: Ah, come on. You You beat yourself too

robert maria: just give lecture. Yeah. I just give a lecture and I give projects and I speed them in teams but I uh don't have

Claudiu Leoveanu-Condrei: much.

robert maria: much of a feedback on uh so yeah

Claudiu Leoveanu-Condrei: Anything. Anyways, you have the most experience in teaching. So, you know, things that work or or don't work.

robert maria: teach students say uh by doing by

Claudiu Leoveanu-Condrei: That's my point.

robert maria: examples demo they like demo life coding things like this because

Claudiu Leoveanu-Condrei: Yeah. Okay. And I'm

robert maria: also it's computer science They they don't like math.

Claudiu Leoveanu-Condrei: being

00:15:57

robert maria: They don't want but I sometimes I put some a little bit of but not too much.

Claudiu Leoveanu-Condrei: Oh, so we said it's a combination of of um um of the

robert maria: So

Claudiu Leoveanu-Condrei: two. Then I I'm fine with that. I will anyways read through it so we can we we can sync. Um maybe this is another good question. Um should we keep the we keep the talking as you said Louis right on the resources or some some channel like that right where we put only this kind of stuff on it

Luiza Corpaci: Uh so for the resources what I thought was more like that being used for like a link dump um not necessarily for like a link where we dump

Claudiu Leoveanu-Condrei: for what? Ah,

Luiza Corpaci: links.

Claudiu Leoveanu-Condrei: okay. Got it. Okay,

Luiza Corpaci: But for discussion probably some of the others maybe because then we just have like the discussion channel

Claudiu Leoveanu-Condrei: fine.

Luiza Corpaci: that's separate from the ones where we send the the

Claudiu Leoveanu-Condrei: I mean,

00:17:04

Luiza Corpaci: resources.

Claudiu Leoveanu-Condrei: I don't know exactly what we h I mean, we have this lecture Q&A I made. Uh, then we have we have the book. Maybe we can discuss here in this channel.

Luiza Corpaci: Yeah.

Claudiu Leoveanu-Condrei: feel free to I don't know if you if you can uh create channels or not but um maybe you can try and and tell me if you can okay I have

Luiza Corpaci: I cannot.

Claudiu Leoveanu-Condrei: to look into that then uh or or tell me what channel do I do we need more and I will make one I try to keep

Luiza Corpaci: You can maybe like do something with roles and then like

Claudiu Leoveanu-Condrei: it

Luiza Corpaci: um give different like rights based on roles and then depending on how much this group goes then you also have control over that.

Claudiu Leoveanu-Condrei: Okay. So, okay. I I will look into this. Okay. Uh good. Then where were we? Ah here at the book. So I'm fine with that. Um we should do it like this since this is a very short chapter.

00:18:11

Claudiu Leoveanu-Condrei: Um I would not consider it like I would consider this discussion being the introduction of it. Uh so that the next week we can directly you know go into um what is dependent type theory and all of this. So uh let me think is there anything else orwise that I wanted to say we kind of touch everything that's important um before we move on to this there is the installation that uh we should all do so there are two ways to use lean and um is this website I will again drop it so that you you know guys where to I mean it's like if you type installation lean you can also find it but I will drop it. So there are two ways to use it. Uh VS code is the most uh robust way. U they they put a lot of work into having a VS code extension that is really really good and I I didn't use VS code in my life too much because I hate it. So I'm not the VS Code user.

00:19:24

Claudiu Leoveanu-Condrei: Uh but I am the neo neoim guy user. So I know I'm using neoim. So I'm I'm having this plug-in installed. Uh but if you guys use any other editors other than neoim or v code, I don't think there is support for or any of that. I mean I'm pretty sure because I I I checked this. So you can only use VS code or neoim for v for limb to have a really good experience with the with the package. and uh good syntax and error and linting and all of that. So I I don't know if you guys are Vimh or VS Code. Um but keep this in mind or Linux user or Mac OS user or or

robert maria: I have the spot and I

Claudiu Leoveanu-Condrei: Yeah.

robert maria: have cursor too. If uh

Claudiu Leoveanu-Condrei: Oh, of course. I mean, everyone has something today. But uh uh just keep this in mind. It's it's only these two. Oh, actually I see now what you mean.

00:20:25

Claudiu Leoveanu-Condrei: uh cursor. I actually didn't see cursor. Sorry. But yeah, it's a neo. It's a it's a VS code fork, so makes sense. Okay, then I don't know. Uh can you I mean it should be pretty straightforward to install this. I I I don't know how techsavvy you are guys, but do you think this is a problem or should we go now through it for the ones that

Andrei Ungureanu: I have it installed so I'm okay in cursor as

Claudiu Leoveanu-Condrei: Okay.

Luiza Corpaci: It's also okay for

Andrei Ungureanu: well.

Claudiu Leoveanu-Condrei: Okay.

Luiza Corpaci: me.

Claudiu Leoveanu-Condrei: Yeah.

Andrei Ungureanu: But I'll try the new VM just to

Claudiu Leoveanu-Condrei: Sorry if we're making So sorry for making this uh I mean I don't know how how techsavvy are.

Andrei Ungureanu: make

Claudiu Leoveanu-Condrei: I I assume all of you are developers. So that's the first thing. Correct me if I'm wrong.

Luiza Corpaci: Yes.

Claudiu Leoveanu-Condrei: Okay.

Luiza Corpaci: As

Claudiu Leoveanu-Condrei: Then then I I would assume you guys know how to set this up.

00:21:24

Luiza Corpaci: in

Claudiu Leoveanu-Condrei: I'm not I'm not spending any more time on this. Cool.

Luiza Corpaci: Yes.

Claudiu Leoveanu-Condrei: Okay.

Luiza Corpaci: Like um a question on this.

Claudiu Leoveanu-Condrei: Please

Luiza Corpaci: Do we want to approach this as in um like um making no assumptions about our understandings and then building up from this or like going a bit from like a top down where we assume a specific level and then count on each other to ask questions when something is

Claudiu Leoveanu-Condrei: Um I think both in a way right I mean I

Luiza Corpaci: unclear?

Claudiu Leoveanu-Condrei: since we kind of follow the book the book has already some an implicit structure but uh the part with the question is up to us so I I can tell you I ask a lot of questions so I'm fine with that.

Luiza Corpaci: Okay,

Claudiu Leoveanu-Condrei: Yeah, I mean we can always debate around questions and yeah,

Luiza Corpaci: thank

Claudiu Leoveanu-Condrei: I I like that. That's how I learn. So I will do that uh by naturally.

Luiza Corpaci: you.

Claudiu Leoveanu-Condrei: Okay. Um so yeah, I will drop this again in um in the in the discord so that you guys have it.

00:22:36

Claudiu Leoveanu-Condrei: But uh again, it should be pretty straightforward. Uh cool. Then if we're still here, there are also some other things that are important. Uh maybe a bit more. So um this is the LIN version uh manager. So Alan, think like the way you compile things in C++ or whatever. Uh this guy is used to um wait a second. So there there is an important package that usually we will probably use it a lot which is called MATL um this one. So this this has to be built um and there is um there is lake they are using lake to build this. So I am not actually sure I mean I don't actually remember like if is if this is required in the first part of the books but I would recommend that you you guys install it um on your machine so that you have uh MATL installed because that there are so MATLI is like a community package maintained by a lot of people it's an open source and we are just trying to formalize the entire mathematics So this matlab is

uh constantly uh advancing and it's used by literally any library that you can think of.

00:24:07

Claudiu Leoveanu-Condrei: Um so there is a standard library which has this uh it has theorems about natural numbers and uh more standard things in in mathematics like uh uh proofs and how logic works and uh stuff like that. And then there is the MATLI which tries to go in more depth about um specific theorems that you might be using. Um so again I will drop this um it should be again pretty straightforward how to set it up. Um I I don't know if you if you have any questions please ask. It's again you just clone this and you build it and that's pretty much it. And there are other probably in VS Code you have different workflows that you can use to set it up set this up more nicely. Um but yeah essentially this is like um you would import um or you would install a package in Python and you just import it and you use it is the same thing. But what's what's nice about LIN is that you only need one import. is saying it's one dependency because everyone is using this you can considering standard if you want so yeah so there is this uh then there are some tools uh that I don't have them right now open uh because I forgot to open them uh but there are some tools that would help you a lot to um check whether or not there is a specific kind of formula or there is a specific kind of um I don't know

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Claudiu Leoveanu-Condrei: like strategy in the um in the library because this library is like huge. It's like it's it's very large. I don't know if I have it locally, but uh let me see. Do I have it? Uh no, I don't have it. No, but uh it's huge. It's like I think it's millions of lines of code or something like that these guys put into it. So, it's a lot. Uh but uh yeah there is a way to search this library. They they build a like a search engine which is cool. Uh and for instance in neoim there is like a package that would um let you search from the neoim and I assume it's something similar in VS code. It has to be something like that. Okay. Um, so this let me see if there is anything else about installation and stuff like this that I'm missing right now. Um, no I don't think there is. But before we move on, is there anything you you want to ask for this

00:27:12

robert maria: me I'm still I still I'm curious about lean compared to other functional programming and and if you want to prove like mathematical theorems what's the difference between this and mathematica but this is not for proving maybe for but other software. What's so special about Lynn? If you have an idea why there's a community growing around this, what's the

Claudiu Leoveanu-Condrei: Well, it's I mean the why is always a combination of different

robert maria: promise?

Claudiu Leoveanu-Condrei: factors, right? the the why is also a combination of how good is your um um sales right it's also combination of are you in the right time at the um in the right place and and all of that so it's always something like this but what kept lean um because this is an old project this is not something new this is lean four so lin four implies that there are lean three and two and one so They every time rewrote it from scratch because these guys are insane. I don't know how else to put it other than you are insane if you constantly rewrite everything from scratch.

00:28:31

Claudiu Leoveanu-Condrei: Uh so they did this because they always wanted I mean this was like a design choice from the guy that invented it which is um uh this guy um Leonardo de Mura. This guy invented um LIIN 4 I mean LIN and he is also the invent the inventor of Z3. I don't know if you guys use V3. I use V3 is an S S&T solver. So um the difference between an SMT solver and a a tier improver is that in an SMT solver you don't have um first order logic and stuff like this. you you you can only do constraint um satisfac solve constraint satisfaction problems. You cannot do uh proofs um of higher orders or anything like that. So that's the difference between solvers of this kind and um a lean. Now what is nice about lean is that it's also a programming language but it's also uh an automated theorem prover which is a very interesting thing. So you can use lin to write programs like literally code you can code in it's a touring complete language so you can code anything in lean um but at the same time you can use lean to write proofs because proofs are just are just programs that's that's what they are there is an important uh concept in mathematics that ties mathematics with computer science which is called curry powered um

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Claudiu Leoveanu-Condrei: correspondence this thing and you can read more about it. It's a quite an old concept. It's developed by this guy Curry. Curry is the guy that um basically um Haskell started from. Um if you know Haskell, Haskell actually it's a language that's very similar to um yeah

robert maria: Yes, I know. I know.

Claudiu Leoveanu-Condrei: so it's very similar to Lin 4 in spirit um only that it was used more like an academic uh fetish by the guys that uh I don't know used to run um very high level mathematical discussion I think in the 80s or I don't know exactly when was that but uh they they Haskell uh to talk their language in a programmatic way and I coded a bit in Haskell at some point and it's and it's actually really cool. It's a it's a nice language and it's very similar to Lin 4 in spirit. But again this this correspondence is the key of all of this is the key of um formalization is at is at the heart of why how what do you mean when you're saying that there is a proof but actually I I I read code right so I read a code but actually this is a proof what exactly do you mean when you say this right so this is fundamental Because this implies that any any proof is a program or any

00:31:46

Claudiu Leoveanu-Condrei: program is a proof. And any language that has dependent type and dependent types um is a way of saying that you cannot program in that language unless you have very clearly defined types in it. So for instance in lin 4, lin 4 is a dependent type uh language and everything has to be typed in it because this language doesn't understand anything outside of types and types

robert maria: Yeah. Is it related to type scripting like uh in uh I forgot

Claudiu Leoveanu-Condrei: are

robert maria: in web uh which language web programming but um something

Claudiu Leoveanu-Condrei: no.

robert maria: with it's not JavaScript I don't know where you have to everything you have to now even in Python you have

Claudiu Leoveanu-Condrei: Yeah. Typescript.

robert maria: it's TypeScript you have to define everything and you is it related to

Claudiu Leoveanu-Condrei: Yeah. So as a as as this idea.

robert maria: this

Claudiu Leoveanu-Condrei: Yes. But but those languages are not uh dependent type theory language languages. It's just any language accepts the idea of a type. So type is

just a way to introduce semantics into your program.

00:32:59

Claudiu Leoveanu-Condrei: That's what a type is. And you can you can make highly abstract types, right? So if you're using Python, you can define the type of a person however you want to be. Um so that that's that's what's nice about types. They allow you to create a semantics for a very tiny subset of your program space. So if you're coding something about, I don't know, I'm giving an example, let's say animals, right? Then you might have a type system for animals and you do inheritance and you do polymorphism and you do all those things that are happening in in programming usually. But Typescript and pro and Python and all these other um languages they are not dependent type languages. A dependent type language is usually one in which you can do theorem proofing because all the types are used underneath as circuits in a way they are just creating. So all the so the compiler of a type dependent language usually has tricks of how to you know wire um from one type to the other so that you reach like a let's say like a like a how to call it like a block or like a unit of code that is self-consistent in itself like an axiom and based on that you can literally um make a chain that when that chain

00:34:38

Claudiu Leoveanu-Condrei: executes then basically it becomes a poof um in your um of your program let's say and that's that's what's so powerful about these things it's what what one of the things that is very powerful is that um for instance again if you compare to Z3 in Z3 you cannot have hoops about um infinite so you do cannot have infinite descent or you cannot have that there's no there's no such idea in in in a Z3 solver about how to handle infinities but in this um in lean you can talk about infinite statements because there are theorems which are already proved that you can use as building blocks for achieving a certain argument. So it's not quite like Typescript. Um so the fact that they are using types um is more fundamental than Typescript or Python uses types because here the type is the it takes the shape of your argument and and you can you cannot program you you will see uh I wanted to show show you something and I'm going to show you now I think it's there. So, I don't know if you guys know this.

00:36:03

Claudiu Leoveanu-Condrei: This is part of the um they recommend this. It's called the natural number game. And this is like a tutorial of how to interact, how to use lean basically. And you and you and you will prove that 2 plus 2 is four. And you would prove um um commutativity and stuff like that. So you if you never coded in in in any functional programming of this kind but especially not even Haskell if you coded in Haskell it's it's different but when you code in in in in Lin 4 every single symbol that is a variable that you write you have to give it a type otherwise you you the compiler would scream back at you and that's a very different kind of thinking because you you will always it's like you would always have to think in advance of how you approach the problem because am I giving this let's say you have two numbers and is it let's P and Q and am I giving P like a rational or am I making P rational so you always have to make a decision of this type when you write lean literally every single variable every single statement is a type underneath and I will also share this uh what what this is teaching you I mean if you this is very addictive so if you spend time through it um it's going to show you the most important um axioms in mathematics so those are the piano axioms uh

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Claudiu Leoveanu-Condrei: which I I don't know if if you know but there is this idea of successor and um the successor of a of a successor and so on and so forth and as that's giving you the integers and because you have the integers then you can recreate literally any any other uh system like rationals irrationals and so on. Uh so this is teaching you like all the theorems of the piano axioms these are called tactics. So tactics are way of applying um a certain procedure to achieve a goal. So I don't know you might you might take two variables let's say P and Q and you I don't know like you do a left substitution or you do like a right substitution or you um decide to reflect or this you make it a tology and so on. So again you will see when you reach a tology for instance it's very weird to think like this. I mean if if you're not a mathematician and you never spend time thinking about math as code or anything or or you just did some math on paper when you try to make the p the math working on a computer in a language like lean 4 it's going to be you would have to prove every single thing.

00:39:04

Claudiu Leoveanu-Condrei: So you will find yourself at some point looking at the statement and try and and say this the thing that but this is obvious and indeed is obvious right but and that's a tautology it's like it's something that is obvious but you should have you have to explicitly say that that is a tautology and that's how you want to prove it. So it has some nice example about autology and how to how to how to use this tactic and uh it has yeah the definitions of the um of the u operations over natural numbers some of them. Um so this is a very nice uh I would say interactive way of uh learning a bit um lean I highly recommend this I don't know if are any questions or if if you want to add something. I don't want to be the only guy that talks.

robert maria: Yeah, I'll have to go. I'm I'm sorry. There's a big scandal with the

Claudiu Leoveanu-Condrei: Oh,

robert maria: kids.

Claudiu Leoveanu-Condrei: no worries. We can Anyways, I think one hour today is fine.

00:40:23

robert maria: Okay. Well, see you the next time.

Claudiu Leoveanu-Condrei: Uh, sure.

robert maria: Bye.

Andrei Ungureanu: Okay,

Claudiu Leoveanu-Condrei: Bye-bye.

Andrei Ungureanu: byebye.

Claudiu Leoveanu-Condrei: So,

Andrei Ungureanu: At the moment, I'm trying to grasp uh as

Claudiu Leoveanu-Condrei: Yeah,

Andrei Ungureanu: much as I

Claudiu Leoveanu-Condrei: it's it's it's very it's it's going to be weird. You will see. It was weird for me as well.

Andrei Ungureanu: can.

Claudiu Leoveanu-Condrei: I mean, it's like you're going through a statement and it's like, what is this? Like, what do you mean? It is obvious or like, of course, it's like this. And but you have to make it explicit, you know? And that that that's going to be a very weird like uh

Andrei Ungureanu: Yeah,

Claudiu Leoveanu-Condrei: step.

Andrei Ungureanu: I guess it's part of the challenge to uh like rewire your brain a

Claudiu Leoveanu-Condrei: Yeah,

Andrei Ungureanu: bit to think uh

Claudiu Leoveanu-Condrei: it it will be a nice rewiring of the brain. I mean I at some points I was looking over uh sil what the hell is those called physics fib maybe li I don't know uh don't remember how it was old uh physics.

00:41:52

Claudiu Leoveanu-Condrei: Oh yeah, this one. So, wait, what the hell was this? Lean and athletic. Okay, cool. Uh this is this is a cool uh project. So they try to formalize the physics in in le and uh this is the guy that that uh is running the the project very nice guy I was looking over some things that he built it's very cool but anyways I'm a so my background is machine learning uh but I learned machine learning from statistics perspective so I'm I'm got good with that so it was here somewhere um something about distribution something something uh I think it was maybe this uh yeah so I I spent I think an evening going through this I obviously didn't understand everything um or most of all of what's happening here but I understand the math that is is here But I was like I mean if you they have comments like this. So I love this. I mean this is like like you you open this stuff and there is there are comments that are very long but they they tell you like why is this like this or like why it's happening this way and not another way.

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Claudiu Leoveanu-Condrei: And this is this is a lot even in lean in the in the main library. If you have I don't know like um check or proposition or what is and um like with capital and uh then if you go to definition to the end there is a lot of there are a lot of comments there like a lot of comments that explain why is it like this and not the other way. So, I spent some time reading this because I I I like physics and I was really curious like how how the hell do you formalize like a distribution in lean formats and I was reading what this guy did. I think it's a Chinese guy and this is like a pleasure to read. It's like it's it's incredible that something you can do something like this. I mean I I wish I I can understand all of this but I don't. Uh but the comments are incredible. So it's like it's like a book really a book with annot

Luiza Corpaci: There's

Claudiu Leoveanu-Condrei: annotations. Um yeah,

00:44:31

Luiza Corpaci: one about time like there's one folder

Claudiu Leoveanu-Condrei: sorry

Luiza Corpaci: called space and time and it goes into from writing

Claudiu Leoveanu-Condrei: I Yeah,

Luiza Corpaci: time.

Claudiu Leoveanu-Condrei: I actually I've read one of the issues that these guys had. I don't know if we can find it, but it was something about second. How to define second? Yeah, there you go. Uh, and this guy abandoned it because it was too complicated to come up with a good idea about

Luiza Corpaci: Oh,

Claudiu Leoveanu-Condrei: what the hell is a second. So, exactly like that's my point is like we all know what is physics, right? So, I mean I played with things in physics. I programmed simulations and stuff like that, but like what does it mean to formalize physic on a computer? like literally to speak about infinities and truly demonstrating a statement that you

Luiza Corpaci: heat.

Claudiu Leoveanu-Condrei: know the pendulum behaves like this for a reason and there is a proof of that and I've seen this like of course like you got to define seconds right you got to define the any any measure that you uh any unit that you have you have to define it.

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Claudiu Leoveanu-Condrei: So then is a question like how do you define a second of um yeah he this guy said that he tries to define it like this and it failed and then he moved the discussion I think this is on tulip or I don't don't know yeah zulip sorry so he's basically saying how the hell do we define a second man like of this is a second basically it's using this atom to define a state a second but like how are you doing this right when you're doing maths and you're doing lean and that's like such a at least to me I mean again I'm not I don't speak their language I don't understand what they do but I understand physics and then I was thinking how even you define a second in you know

like there was you don't have this kind of question like how do I define a second in lean that's that's a very odd question to ask you will only ask it if you kind of work on this kind of project which to me is like it's very interesting that there are guys like this that are building this kind of stuff and it's very interesting to read and I my hope is that as as we progress through the book and we get more knowledge and we get familiar with all this stuff I we can go like each one of us probably I mean it's interested to learn this for a reason or or

00:46:58

Claudiu Leoveanu-Condrei: the other but like I would definitely be very curious like what these guys did in more depth for instance because I like physics so to me This is like a band physics and it is much easier to parse obviously than other things. But yeah, there are cool stuff like this. And one other thing that I wanted to show you uh which is a very good resource um and you should definitely use it. Let me go to lean for language uh this one. So you can always take like a repo whatever repo you find and you can always um go here and replace GitHub with deep wiki and if it's not indexed already you can ask to index it but uh yeah this these are some guys uh from cognition the guys that I don't know how much you follow this scene but these are the guys that uh I'm doing this for my job. So these are the guys that build Devon which nobody uses or I don't know who the hell uses Devon but anyways who is Devon but they build this stuff and this stuff is one of the coolest thing most useful things I've ever seen uh because there's a lot of hype and BS about AI but this is actually something useful and good.

00:48:24

Claudiu Leoveanu-Condrei: So this is like documentation u for your um yeah build with with rag underneath they are using rag and some models and so on but that's not the point. The point is that this is great to parse complicated um projects. So if you would like to know um more about lean 4 as as as as people that don't understand this kind of stuff in in any depth, but you are still the kind of people that read code or reason more in code better, then this is a great resource because first of all you can ask things. Um but second of all, you can literally go through the um through the code in a more organized way because it has diagrams and it's telling you like okay then like how the hell is this all of this related and um for me this is like very helpful because I'm I'm also I'm I'm more of

a visual guy and I and I like I like things like this that um and unpack a lot of uh dance material. Um, but I I at least I can visualize it somehow more easily.

00:49:40

Claudiu Leoveanu-Condrei: Uh, but it it's very good. So, I recommend this um as well if you want to read more. And literally any GitHub um has one or you can index it if it doesn't exist, you can request um to index it. Uh but yeah, this is u I will also put this as a resource because it's going to be useful. Um yeah, anything from your side, guys?

Andrei Ungureanu: Not at the moment. I think

Claudiu Leoveanu-Condrei: Okay. Uh, did somebody leave or somebody joined? I I don't

Luiza Corpaci: I left and joined my internet.

Claudiu Leoveanu-Condrei: know.

Luiza Corpaci: Something happened with it and I

Claudiu Leoveanu-Condrei: Ah,

Luiza Corpaci: disconnected.

Claudiu Leoveanu-Condrei: okay. Cool. So, uh, what I asked before I think you join or before you left it was if there's anything else we should be discussing now or we we Okay.

Luiza Corpaci: Uh yes I have a few questions.

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: So one of them would be about the the writing of the code. So like what's the approach we want to follow?

00:50:58

Luiza Corpaci: Do each of us uh like write code for all the exercises or do we write the code in the the meeting or just generally about that like how how do we approach that?

Claudiu Leoveanu-Condrei: H I mean I

Andrei Ungureanu: Well, if the code is part of The book we should do

Claudiu Leoveanu-Condrei: think and we should do

Andrei Ungureanu: beforehand.

Claudiu Leoveanu-Condrei: it. Yeah, I agree. Yeah. And we can discuss the code in the session.

Andrei Ungureanu: No.

Claudiu Leoveanu-Condrei: Yes, with the person who who's a guide um yeah it can show

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: us or for instance if if I would I don't know let's say we are at chapter three and there is something about I don't know let's say rotate left and rotate right right

Luiza Corpaci: So

Claudiu Leoveanu-Condrei: uh so there's no example about rotate left and rotate right well I'm going to do it right I'm going to try to make an example for rotate left and rotate right and try to show it so that we get the gears for okay where would I use rotate left and rotate right?

00:52:05

Claudiu Leoveanu-Condrei: Why is

Luiza Corpaci: okay.

Claudiu Leoveanu-Condrei: that

Luiza Corpaci: And in terms of keeping track of the code in the GitHub, should we all have our own branch and then the person that does the guiding on that session uh

Claudiu Leoveanu-Condrei: no we I invite you you just have to give me the GitHub

Luiza Corpaci: like

Claudiu Leoveanu-Condrei: uh your GitHub handle and we just push to main. There's no this is not something it's just for tracking. We don't care about there's no needing for branches or or you mean in general like when you're working on it? I mean when you work like I would keeping on on your machine and we just push it afterwards, right?

Luiza Corpaci: Yes.

Claudiu Leoveanu-Condrei: Because we only push to GitHub after we finish the session,

Luiza Corpaci: But

Claudiu Leoveanu-Condrei: right? To have the what happened or or you or you mean before if you do something beforehand.

Luiza Corpaci: to some degree mostly the question is about like per session like do we have all the code saved or do we only have the code of the person guiding saved?

00:53:06

Luiza Corpaci: That's the the gift of the

Claudiu Leoveanu-Condrei: H. I mean,

Luiza Corpaci: question.

Andrei Ungureanu: I think the guy should push the his code in the main and we follow that code and I'm going to keep my own repo as well for what I do. So I think the two can be separate like

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: Yeah,

Andrei Ungureanu: that.

Claudiu Leoveanu-Condrei: I mean we can fork this repo and then when we push we push into this one. Sure. But I would try to keep it simple. I would just um I would just invite you as contributors or I don't know how we do it. Um uh but the point is that this is how I would do it. So if if you have something to present or the guide right the guide of the session has some code um then I then I would rather let the guy um walk me through the codes when they are presenting the code right but in the same session I might also have a code right because I wrote something.

00:54:18

Claudiu Leoveanu-Condrei: So then I'm going to present you my part of the code and then once we finish the session all this goes to the repo and then we can um uh I would say um but I don't know like go and or I don't know you want to use the code or to learn more about it. That's I think that would work better than having also the code a priority to go through it. I think already having the chapter of the book go through um which has already code in it. I think that's enough for that for the guide in my opinion is is is better to let

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: the that person uh go to the code and also again if we have our code I mean we discuss in the session if I if I also have some code right it's u I I can if we have time I can show it right or I if if not then I can just push it to the uh people.

Luiza Corpaci: Yes. And we can also use discord like throughout the week to

00:55:23

Claudiu Leoveanu-Condrei: Yeah, of course. I mean, of course. Yeah,

Luiza Corpaci: discuss

Claudiu Leoveanu-Condrei: I I'm still thinking about this stuff. I don't know exactly what should we do to set this up. Um, do we need this? I don't think we need this. I mean, I know this is the community of the lean, so uh I don't know exactly

Andrei Ungureanu: What do you mean?

Claudiu Leoveanu-Condrei: what Uh,

Andrei Ungureanu: I'm not sure what you're talking about.

Claudiu Leoveanu-Condrei: so can you see my screen?

Andrei Ungureanu: Yeah.

Claudiu Leoveanu-Condrei: Yeah.

Andrei Ungureanu: Yeah.

Claudiu Leoveanu-Condrei: So this is like the official community of the of the lean people.

Luiza Corpaci: Yes.

Claudiu Leoveanu-Condrei: Literally everyone is here from that guy there.

Andrei Ungureanu: There's like a forum.

Claudiu Leoveanu-Condrei: Yeah, exactly. It's a forum based um um like I don't know. It's forum. Yeah, it's a forum based communication channel. So this is Yeah, there's a lot of things happening here. is literally every single person that is doing lean professionally or it's really good is here.

00:56:19

Claudiu Leoveanu-Condrei: So I don't know how we should whether we can make a channel here or like I don't know exactly how it works. Um whether you need to add or you make a request or I don't know I don't want to spend too much time on this stuff.

Luiza Corpaci: can do that if you want. I'm also in that. I'm in there.

Claudiu Leoveanu-Condrei: Um,

Luiza Corpaci: I hadn't done that already. I think one either like we should ask one of the mods to do it or we need to get mode right, but I'm not sure if someone would give that to us at this point.

Claudiu Leoveanu-Condrei: yeah. Then if you want to look into this, I'm fine. We can have a group for us. I don't even know how to lean mode. Yeah. call it ask for limos but I don't think there

Luiza Corpaci: Okay. Yes. And then maybe Oh,

Claudiu Leoveanu-Condrei: is

Luiza Corpaci: sorry. And

Andrei Ungureanu: Why would we need this to to be present here on this forum?

00:57:08

Andrei Ungureanu: I

Claudiu Leoveanu-Condrei: I don't know in case go ahead

Luiza Corpaci: there's lot of expertise like my thought would

Andrei Ungureanu: must

Claudiu Leoveanu-Condrei: sorry

Luiza Corpaci: be that there's lots of expertise there and if we want to have other people join then that's going to be like a good place for them

Claudiu Leoveanu-Condrei: yes it's like it's it's

Andrei Ungureanu: Okay.

Luiza Corpaci: to join from.

Andrei Ungureanu: Yeah. Yeah. Fair enough.

Claudiu Leoveanu-Condrei: it's so for instance um I might have a question right so I know this guy is here so I can just tag the guy but in the discord I cannot tag this guy because he's not on our discord so that's more like the uh here for instance if we have the

Andrei Ungureanu: Mhm.

Claudiu Leoveanu-Condrei: channel then I can we can make this like slightly more official right so I can I can just tag this guy in the in our channel and just ask him if you want give us a talk. That's I would say more uh we're already part of the community, right? So yeah, Louisa, if you want to look into this, I I would be happy with it.

00:58:05

Luiza Corpaci: Okay,

Claudiu Leoveanu-Condrei: Yeah.

Luiza Corpaci: then I can have that uh on my

Claudiu Leoveanu-Condrei: Okay. Yeah. I mean,

Luiza Corpaci: res

Claudiu Leoveanu-Condrei: we should always be part of the community as much as possible, obviously. Uh yeah. So that would be it from my side right now at

Luiza Corpaci: Um, I would have another question for how we decide who guides which.

Claudiu Leoveanu-Condrei: least.

Luiza Corpaci: Do we do it on a like week by week basis or do we, I don't know, maybe set an Excel file and then everyone can put their names in front of the chapter or something like this.

Claudiu Leoveanu-Condrei: Um like you mean in

Luiza Corpaci: Yeah, sort of. cuz if we like currently we know the chapters there's eight

Claudiu Leoveanu-Condrei: advance.

Luiza Corpaci: chapters without considering the next steps and maybe also without considering like the first one. So there's seven chapters and and then if we know who does which

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: then also like

Claudiu Leoveanu-Condrei: Yeah. I mean Yeah, we can. I'm fine with that.

00:59:07

Luiza Corpaci: we

Claudiu Leoveanu-Condrei: But I don't know if Excel or can I mean can can we do like um can you do tables in Discord? I don't think so. Right.

Luiza Corpaci: can also have it in the readme on the GitHub.

Claudiu Leoveanu-Condrei: Of course that that's I would do it. Yes. There as Yeah, I would do it there. If we can if we can do that then I would do it there.

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: Yes. I would try to keep everything on GitHub. It's it's the easiest in my opinion.

Luiza Corpaci: Yep. I I like the idea.

Claudiu Leoveanu-Condrei: Okay, cool. So then I think I have your email addresses. So we we're going to keep then the meats. Um we keep discord for chat. Um unless wait what the hell is happening? Ah oh my god I clicked on a on a channel something started to the staging I think. Anyways, uh we keep the discord for chat and then the meets for the the sessions,

01:00:19

Luiza Corpaci: Yes. And I forgot,

Claudiu Leoveanu-Condrei: right?

Luiza Corpaci: did you say that this is going to be a weekly or bi-weekly uh reading or
Andrei Ungureanu: Okay.

Claudiu Leoveanu-Condrei: I mean,

Luiza Corpaci: discussion?

Claudiu Leoveanu-Condrei: I would go with weekly because it's going to keep us warm.

Luiza Corpaci: Okay. So, next Thursday, same time.

Claudiu Leoveanu-Condrei: Mhm. Yeah. So we should so ideally we should try to be in the two hours bucket uh because it's still going to be late. So I I don't know. Um I hope I think we don't need more than two hours, right? And I think somebody should check keep check of the time because it could be that we derail and we we got to go for hours. So I can do that. Um I can I can keep time but in my opinion I think we don't need more than 2 hours in general. So 911 I it's looks it's decent for me. I don't know if you Yeah.

Luiza Corpaci: also but

Andrei Ungureanu: Yeah, it's

01:01:18

Claudiu Leoveanu-Condrei: Okay. Then we keep it like

Andrei Ungureanu: okay.

Luiza Corpaci: then yes but like a thought that I had is that if we want to keep track of time we should also

Claudiu Leoveanu-Condrei: this.

Luiza Corpaci: have like a way of knowing what the sort of agenda like a draft agenda should look like maybe to to I don't know have half an hour for like the general chapter discussion and open topics and then looking into the code and then leaving at the end some like open discussion or something

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: like this just as a broad stuff but because otherwise otherwise we would just figure out when the

Claudiu Leoveanu-Condrei: Yeah.

Luiza Corpaci: end is without like a minimal structure

Claudiu Leoveanu-Condrei: No, no, no. We should No, no. We should have a structure beforehand.

Luiza Corpaci: basically.

Claudiu Leoveanu-Condrei: Yeah. But we can we can discuss it in the on discord before before the before the

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: meeting before the session. Sorry. Yeah, I I I agree.

Luiza Corpaci: Mhm.

Claudiu Leoveanu-Condrei: I mean we should be as organized as possible considering we have two hours for a session.

01:02:15

Claudiu Leoveanu-Condrei: So

Luiza Corpaci: Yeah.

Claudiu Leoveanu-Condrei: yeah.

Luiza Corpaci: But this is nice because this is like an emergent organization that comes up.

Claudiu Leoveanu-Condrei: Yeah. I mean I'm curious like how it's going to you know converge and how how it's going to happen. But I mean I I as I said I like learning with other people. I I didn't do this in a while. I actually learned with Robert in the past. Um we we kind of studied machine learning at session and stuff like that. So yeah, I mean I like this. And when when Nelson joins us, he's um I think from from all of us,

Luiza Corpaci: Nice.

Claudiu Leoveanu-Condrei: he's the he has a PhD in uh in in symbolic logic and stuff like this. So he's really he's really into this stuff. He likes it.

Luiza Corpaci: Oh, wow.

Claudiu Leoveanu-Condrei: So he's going to be Yeah, he's going to be very very very a very cool presence uh in the in the sessions.

Luiza Corpaci: Do you think it makes sense to add like at the beginning of the next session or maybe at the end like a mini spot where we each give a mini description of like what brought us to Lynn and like what's the goal of using it because then we also have a good map of like who's doing what and for which reason and also save connections and so

01:03:33

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: on.

Claudiu Leoveanu-Condrei: Yeah, we we can do that at the beginning of the next session. Yes, like five minutes or something like that.

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: Yeah, just a habit. Yeah, we we can do that. I'm fine.

Luiza Corpaci: Okay. Nice.

Claudiu Leoveanu-Condrei: Cool. Then I think we can close this one if there's not anything else. Uh, I'm going to put

Andrei Ungureanu: Uh but I think we we should talk about uh what we will go through

Claudiu Leoveanu-Condrei: it.

Andrei Ungureanu: uh next week from the book.

Claudiu Leoveanu-Condrei: Uh, I I thought we decided to do that on Discord, but I'm fine.

Andrei Ungureanu: Ah, okay. I missed that.

Claudiu Leoveanu-Condrei: Yeah,

Andrei Ungureanu: Yeah,

Claudiu Leoveanu-Condrei: because Louisa Louisa said to have a like a structure before the

Andrei Ungureanu: maybe on Discord.

Claudiu Leoveanu-Condrei: session.

Andrei Ungureanu: Yeah, it's better to have it written

Claudiu Leoveanu-Condrei: Mhm.

Andrei Ungureanu: somewhere.

Claudiu Leoveanu-Condrei: And actually we obviously should have this ASAP, right? Um I mean we we should do it this these days.

01:04:48

Claudiu Leoveanu-Condrei: Let's say to always is to have it um by Friday. Is this fine? Because then we have at least we know what's going to happen up until the next Thursday. So it's like six days in advance.

Andrei Ungureanu: Yeah. Yeah.

Claudiu Leoveanu-Condrei: Okay,

Andrei Ungureanu: Perfect.

Claudiu Leoveanu-Condrei: then let's let's let's decide up until tomorrow what what is going to be the next week.

Luiza Corpaci: Okay.

Andrei Ungureanu: Okay.

Claudiu Leoveanu-Condrei: Okay.

Luiza Corpaci: And who is going to guide it?

Claudiu Leoveanu-Condrei: Yeah. Who's going to get it? Yeah.

Luiza Corpaci: And then this information could also go like in the in the calendar invite description as a sort of just to to have it like easily accessible then for everyone.

Claudiu Leoveanu-Condrei: Well, I mean, do we need that? I I don't know how to do this. This has to be manually done, right? And I I it might be that I forget

Luiza Corpaci: Then we can do like a double accountability sort of thing like if you forget I can keep it as a as a backup sort of putting it

01:05:54

Claudiu Leoveanu-Condrei: because if I'm making a meeting now weekly, then that's going to be already scheduled,

Luiza Corpaci: No.

Claudiu Leoveanu-Condrei: right? It's going to be already appearing every week. So I will definitely forget that I have to add that unless I mean, I I don't remember right now, but I probably will remember at some point. I know

Luiza Corpaci: So when I when I had the idea for the structure,

Claudiu Leoveanu-Condrei: myself.

Luiza Corpaci: I had it more like a sort of a template for the u the meeting not specifically like very in detail as in what is it going to be in terms of the next chapter but more like some

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: slots for time such that when we keep track of time we know what should we compare the time to. go more like in that direction. But then Andre was having a more like pragmatic question as in what do we prepare for next week? Um and that to be fair now I don't really know like

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: besides going through the chapter and having the setup done and doing some of the exercises I'm not sure how structure would look like.

01:06:59

Claudiu Leoveanu-Condrei: Um yeah. So again we kind of rule out right that we go through the reading during the session right. So that's that's no longer the case. So we kind of talked about that we we try to read in our time the chapter before or like let's let's not call it the chapter okay because the chapter as we said could be split we might

decide that okay this is maybe too long or too tedious to do it in one go so we might split it in two or three sessions. Um so we decide I would say we decide beforehand what is the material and who is the guide right so this we have to do up until Friday and the guide is kind

Luiza Corpaci: Yep.

Claudiu Leoveanu-Condrei: of either so either we go full code and we do the examples in the code and we reference the book where we need to reference the book uh I mean I would always try to reference the book to some extent right I mean if there are questions Then if we all read the material, then at least there are four there are five brains that already read the material.

01:08:16

Claudiu Leoveanu-Condrei: So at least one of those brains I assume that they already know or or or they they know where it is in the chapter this information or maybe there is that information is not in the chapter, right? So then it's like it becomes u either like a yolo uh how the hell do we answer that question? What is that? That could be part of the next session or something or or have like a interlude or I don't know how to call it. But this this to me is more like a free form. This is part of that free form emerging structure because I also don't know how it's going to be. I I think what's important is to decide the um up until tomorrow to decide whether we do the entire chapter too um and who is the guide and um we should read the chapter or the material that we have and do the coding each one and then the guide just just goes through the code. That's what I would do. it would go through the code and pick the um let's say if there is no codes right usually there is code but let's say there is no code so what I would do if I would be guide is exactly what I showed you before so I I see that it's talking about left and right

01:09:33

Claudiu Leoveanu-Condrei: so I obviously have the inherent question of what the hell is left and right if they don't explain to me or they don't show me code right they don't tell me anything about left and right then I'm going to try to make an example about left and Right? I could two things might happen, right? I could fail to do so or I or I can have an example. If I fail to do the example, then that's actually a good opportunity for us to learn because why did I fail? I mean, I must have missed something or I must have um

not understood something or actually having an example for that concept is quite complicated for the stage we're in and that's why there was no example for it. So, you see my point? I I I think this is all emerging my

Luiza Corpaci: Mhm.

Claudiu Leoveanu-Condrei: opinion.

Luiza Corpaci: And you think that it's best to skip the getting to know Nolan chapter which

Claudiu Leoveanu-Condrei: Yeah, we kind of did it today.

Luiza Corpaci: like

Claudiu Leoveanu-Condrei: Uh I mean you can read it because again uh as Andre said the rabbit hole.

01:10:35

Claudiu Leoveanu-Condrei: So if you if you go if you stomp into a calculus of construction, trust me,

Andrei Ungureanu: Mhm.

Claudiu Leoveanu-Condrei: that's a pretty deep rabbit hole. It's a fun one. I mean it's a lot to read but again I think so anything that's it's up to each one's time right I don't know what you guys do in weekend or what plans you have or stuff like that but everyone is feel it can read as much as possible and share with others. So I I don't know how is it. How is

Luiza Corpaci: I I mean generally yes,

Claudiu Leoveanu-Condrei: it?

Luiza Corpaci: but like it seems that the getting to know Lan chapter still has a bunch of things like it goes a bit more in detail on the types which today feel like an important discussion like in the summary there's like a recursion thing which could be interesting to figure out how Lin deals with it or or stuff like that which getting the chapter is

Claudiu Leoveanu-Condrei: You're talking about the introduction, right?

Luiza Corpaci: called getting to know Lynn which is the one before her world.

01:11:37

Luiza Corpaci: Ah okay so no you meant the introduction.

Claudiu Leoveanu-Condrei: I

Luiza Corpaci: Okay that's what so you mean to start from getting to Norin.

Claudiu Leoveanu-Condrei: uh

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: Yes.

Luiza Corpaci: So when you said chapter two, I just looked at the number and thought we skip the number one thing.

Claudiu Leoveanu-Condrei: Ah, no, no.

Luiza Corpaci: My fault I misunders

Claudiu Leoveanu-Condrei: I I I just meant that the introduction like um so basically I'm talking about

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: just just so that we're clear I was talking about this stuff I'm showing my screen uh so I was I was talking about this stuff so there is this

Luiza Corpaci: Okay. Yep.

Claudiu Leoveanu-Condrei: uh so this is the book these are the chapters there are 12 chapters and I I meant this one so I I treat that this this is something that we did today because it's mostly Um, I mean, I did this myself. Um, but there's not much other than this, right? U and that's what I meant with the rabbit hole. So if you read this stuff, there's a lot of things in here obviously which are all interesting.

01:12:42

Claudiu Leoveanu-Condrei: So it it could be a very deep rabbit hole, but it's like an introduction, right? He's telling you why are we doing this? Like why why is why this? It's basically what Robert asked more or less. Um so I would I would read this myself. Uh what you should read this obviously but I will start with chapter two and we should decide um up until tomorrow how much we do from basically nine sub chapters right because this feels like it's quite long so I I think this should

Luiza Corpaci: Okay. Okay. Then I had another I had another misunderstanding cuz in the discord

Claudiu Leoveanu-Condrei: split

Luiza Corpaci: uh server uh we have in the in the thread we have the lin lang book which

Andrei Ungureanu: Yeah, that's what I was about to ask.

Luiza Corpaci: goes through functional programming.

Andrei Ungureanu: We have it's a different book on the

Claudiu Leoveanu-Condrei: Oh, I I might Sorry. I might have uh put the wrong book link.

Luiza Corpaci: Yeah.

Andrei Ungureanu: link.

Claudiu Leoveanu-Condrei: Sorry.

01:13:37

Claudiu Leoveanu-Condrei: I I meant this. Where is it? I meant this stuff.

Luiza Corpaci: Okay.

Andrei Ungureanu: So where are the two

Claudiu Leoveanu-Condrei: So this is uh so again the lean 4 is both a programming

Andrei Ungureanu: books?

Claudiu Leoveanu-Condrei: language and a theorem improving. It it has like two purposes.

Andrei Ungureanu: Which that I have the books,

Claudiu Leoveanu-Condrei: So you can write normal codes.

Luiza Corpaci: Hello.

Andrei Ungureanu: one book for each

Claudiu Leoveanu-Condrei: Uh yeah because one is teaching you more like the language aspect of it and the the other one is teaching you

Andrei Ungureanu: purpose.

Claudiu Leoveanu-Condrei: more how to use the language to basically uh prove things like how to prove that 2

Luiza Corpaci: Heat.

Claudiu Leoveanu-Condrei: plus 2 equals 4 for instance that you won't do with the language with the language you do things like I don't know like like you would do with Haskell if you know Haskell like literally code things in the language I mean we can

Andrei Ungureanu: Yes, you replaced the book with this one,

Claudiu Leoveanu-Condrei: obviously reference and it said,

01:14:35

Andrei Ungureanu: right? Because this is the one we're actually

Claudiu Leoveanu-Condrei: "Yes, I I this is what I even shared." Yeah, sorry.

Andrei Ungureanu: following.

Claudiu Leoveanu-Condrei: I might have uh I just Google and didn't look. Uh probably that's why. So, I meant this. I mean, let me put it here because it's the programming

language and this is the um let me This is the uh automated theorem proving. So I meant this this stuff.

Luiza Corpaci: Okay, thanks. Nice.

Claudiu Leoveanu-Condrei: Yeah. Okay. Yeah, sorry about the confusion. Good

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: question

Luiza Corpaci: Then then that's that's clear. I when I was looking at that I had like yeah based on what I was seeing in the other book but the other books looked cool too actually like I skimmed a bit through it.

Claudiu Leoveanu-Condrei: again. It's it you still use lean 4. Nothing changes. Um but this is more aspect of formal verification like where you would use the language right it's like a subset of what are you doing actually with the language it's like the use cases uh this is the mathematics of it uh this one this one where is it they close it right here so this one is the functional aspect of it like how do program stuff in it.

01:16:31

Claudiu Leoveanu-Condrei: It has nothing to do with math that much. I mean, it's telling you probably about math. I didn't look into this, but it's it's functional. It's like defining you see hedgehog and deer and snail. This is not math, right? This is programming. So, yeah. Um, again, you can use both. You can use both uh for both purposes this language.

Luiza Corpaci: Okay. Well, that's going to be fun because it's I hope the learning step is steep. The learning um word learning thing is deep

Andrei Ungureanu: Uh but do you know uh if anyone uses it just for like uh huskell replacement? So just for the programming part

Claudiu Leoveanu-Condrei: I I literally know nobody. that does it for this? No,

Andrei Ungureanu: I mean right so it's uh

Claudiu Leoveanu-Condrei: no, it's mostly for this. uh it's it's how do you how do you do formal verification in um in lean

Andrei Ungureanu: yeah

Claudiu Leoveanu-Condrei: for but again we can we can reference things like if we okay how

01:17:38

Andrei Ungureanu: Okay.

Claudiu Leoveanu-Condrei: do you do for loops right or stuff like that I mean I don't know why we need to do to know that uh for our use case but that's what you would learn here it's like is that I mean for me a programming language is like I don't know if I want to know rust right I read about rust I mean to me that's not something. It's just a I don't know like a um

Andrei Ungureanu: Yeah, it's a tool. It's not u It has to be a tool for something,

Claudiu Leoveanu-Condrei: Yeah, it doesn't tell me like how Yeah,

Andrei Ungureanu: not necessarily the main scope.

Claudiu Leoveanu-Condrei: exactly. Exactly. Yes. So, I would program in LIN 4 to let's say do advent of code, right? You can do that. You can use LIN 4 to do advent of code if you if you if you guys know advent of code.

Andrei Ungureanu: Yeah.

Claudiu Leoveanu-Condrei: um fine fine sure we can do that but I'm not interested in that right I mean that's that's just a normal language aspect of the thing but this is this is the cool stuff of it I mean this is the proving part like how do you use the programming language to prove things that seems more in I mean this is at least from what I know this is what people are using it for all the pro all the projects are um yeah to prove that a

01:19:06

Claudiu Leoveanu-Condrei: certain thing is um correct in mathematics which they use this for I think but

Luiza Corpaci: And what's next?

Claudiu Leoveanu-Condrei: from uh so I think there are sharing share points

Luiza Corpaci: Sorry.

Claudiu Leoveanu-Condrei: of I see here that there is yeah there is like an interlude you see so this this tells me that The interlude is like um I'm not going to tell you what exactly does this mean. Um for this you need to read this. That's what it's telling me. But yeah, it it's teaching you basically the programming language itself. But the but the the you cannot do with the programming language. I mean if you learn this you won't know how to do this. uh we just know how to code in lin for as you would code in I don't know Python or whatever in a functional language. So you wanted to ask

Luiza Corpaci: Yes, I I'm curious about the limits of using formalization.

Claudiu Leoveanu-Condrei: something.

Luiza Corpaci: Like if people are attempting to formalize physics and stuff like time then like where do we reach the limits of this?

01:20:20

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: So like there's there's stuff like g but but then like more practical stuff I don't know like what's the the most complex thing that is formalized or stuff like this.

Claudiu Leoveanu-Condrei: I mean the most complex thing that I know about that is formalized is um a field medals result from um from a woman. I think she's Ukrainian. Some sphere packing thing. uh sphere packing uh bl um yeah whatever I don't know what I have open here but uh is there any ling here so sphere packing formal there you go this this gu is from matt inc so this guy proved this thing in in uh gaus or what was it called I think it was called gaus right yeah us. So this is the a very complicated statement to prove in um in four. So, somebody took the equivalent of the Nobel Prize in math for this and they formalized it and apparently is 200,000 lines of uh link to prove this. Yeah,

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: that's that's why you need a that's why AI and all of this took uh you know took took

01:21:48

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: off right now because you can do 2,000 lines of um fear packing formalization right now which was not possible before I would have writes 200,000 lines of code and it's still saying

Luiza Corpaci: So that also go into the like auto formalization sub field or direction.

Claudiu Leoveanu-Condrei: yes. Yeah. If Yeah. If you're using agents then it's going to be auto is a part of auto formalization.

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: Yes, like Gaus and there are others like Aristotle and Axiom and I can show you stuff like that when we if we reach any point about that because I I even I don't know if you know but I have this u I'm having this is my project that I'm working on uh often uh and I added uh quite recently uh formal verification to the to it

and I have two modes I can use lean local and I can use um Axiom which is like a cloud provider for lean at scale and yeah we can do cool stuff

Luiza Corpaci: That's

Claudiu Leoveanu-Condrei: once we learn more about it.

01:22:56

Luiza Corpaci: cool.

Claudiu Leoveanu-Condrei: Um but yeah where was it? So yeah again I think let me remember where was that? So as a resource uh so if we go to matlib you can see exactly what was already formalized. Um I don't need this. Where is that nice page? Maybe this core tactics. Did they change the page? H how was it called? Mat lily. Um how should I search for this? Um let's say I don't know group theory whatever maybe let's try Google. Ah this is it I think. Yeah, there you go. So, this is basically Matlib, which is that library that I was talking about before. And this this is where a lot of people are spending time to add things to it, right? So, they just keep adding stuff here to grow the to grow the knowledge of the maths that we have as code. So as you can see here there's a lot of like all of these things you we can use in our um in our proofs.

01:24:44

Claudiu Leoveanu-Condrei: So I mean obviously you have to know right that there is a left inverse more left div right. So we need to know that there is that such a thing and that's what this this is what that search engine is for to search this kind of stuff um to find okay is there anything like that and I I spent some time here and um where was that uh maybe here you see so this is basically everything that is already in here so what you see here is already to some extent either started or is in um actually I would say let's say fully matured. So for instance I know there is not a lot of homotopy type theory in MATLIB. I look for this at some point and that's not so much um but this is what was right until right now this is what we have in lin for to work with so more or less problems in this area can be formalized in lin 4 using this mat li um package that's what I was saying all every single library like that phys those physics guy probably if he wanted to prove a second, right?

01:26:09

Claudiu Leoveanu-Condrei: He probably have used Matlib and probably have found some I don't know like something from here that he might be able to um uh use in his code because this is like not starting from scratch. It's like using a library to import a function that is already written by some other guys. Um so yeah, but I I will share this as well. So you literally have everything like how what is the bjection what is like it's everything formalized here from piano actions you can literally find like what

Andrei Ungureanu: Okay.

Claudiu Leoveanu-Condrei: is plus or minus how is defined

Luiza Corpaci: Okay, thanks for the intro or like the the general overview. you

Claudiu Leoveanu-Condrei: matrices I know I know for a fact there are people that used um lean to formalize neural networks. So, and I've seen that a while back. So, it had support even then.

Luiza Corpaci: that that's kind of like one of the directions where I'm trying to go into like the combination of formal verification and mechanistic interpretability.

Claudiu Leoveanu-Condrei: Yeah, that's a cool one.

01:27:25

Claudiu Leoveanu-Condrei: That's Yeah, that makes sense. For me, it's about uh I mean, what I like about this is what why I want to learn more is is because I I'm interested in how do you verify all the slop that the agents are putting out, right? So, everyone is is doing code, but it's full of bugs and all of that. So, my angle is more from security. So,

Luiza Corpaci: Yeah.

Claudiu Leoveanu-Condrei: I'm interesting in vulnerability research and stuff like that. There's a lot there in my opinion that is cool at least for me and physics depending on how much I learn this stuff and how how how much I can understand it. U because I'm pretty sure it's going to be complicated from a point onward. Uh but it's still code and I like that it's code. Um yeah. Uh I don't know. I I'll probably spend some time in the physics one because I like physics but for me it's mostly security related. So as

Andrei Ungureanu: So you want to auto automate the AI

01:28:24

Claudiu Leoveanu-Condrei: a

Andrei Ungureanu: code uh checking.

Claudiu Leoveanu-Condrei: Mhm.

Andrei Ungureanu: That's like the goal.

Claudiu Leoveanu-Condrei: Yeah. something like that. Yeah. Yeah.

Luiza Corpaci: What

Claudiu Leoveanu-Condrei: Or specific protocols from from it like I don't know like you want to you want to verify certain aspects of I don't know let's say if there is something like an MCP right so MCPS are everywhere. So like formalizing that the there is a handshake protocol right between um your agent and the word is something useful. So something like that that that's what I what I like about learning this for me.

Andrei Ungureanu: Yeah. Yeah. It makes a lot of sense in the AI hero.

Claudiu Leoveanu-Condrei: Mhm. Yeah. I I mean if I mean who knows who knows what we'll be working on one year from now spawning from this I don't know but yeah that's part of the journey.

Luiza Corpaci: Uh I I have a question because I saw that like Europo and the project you mentioned before was called symbolic AI and like one of the things I'm currently sort of generally stuck

01:29:35

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: on is on something like a semantic diff checker.

Claudiu Leoveanu-Condrei: I actually did that.

Luiza Corpaci: Um,

Claudiu Leoveanu-Condrei: I implemented exactly that at some point.

Luiza Corpaci: Hit

Claudiu Leoveanu-Condrei: I had the sim. Yeah. Yeah. We had some. So we work with uh it's it's mostly we have a lot of like uh

Luiza Corpaci: it.

Claudiu Leoveanu-Condrei: repositories. I don't know exactly what you can see. Let me see. This is should be only the public ones, right? Yeah. So we have two 29 depositories where we kind of have a lot of stuff here that we tried working on them and so on. Um but uh yeah we actually had a pro we have a project that is ongoing. We we're like a combination of research company and applied um so we had this exact I had this

thoughts six months ago or something like this. I wanted a semantic diff checker and the way we build it was through um use logic because this guy Nelson uh he's working at my company.

01:30:46

Claudiu Leoveanu-Condrei: So the way we thought about it is that we need some sort of like um you need a compression scheme and then we thought that we can use logic as a proxy and if you use logic as a proxy then you the the the logic that you obtain back is um your compression of the natural language text and then you do the semantic diff uh between um in the logic space of the program Like for instance, there is something called uh answer set programming and we're using a lot of this

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: stuff. Oh god, what? Sorry for the I forgot that we are on the so let me try here answer set programming. Yeah. So we use this kind of stuff to do the semantic

Luiza Corpaci: Okay, that that's interesting.

Claudiu Leoveanu-Condrei: div.

Luiza Corpaci: Currently I have like a more naive sort of dealing with it which is mostly also doing a sort of compression towards something that's like that follows some structured atomic rules format and then and then from that like using that as an intermediary representation and my interest is mostly

01:32:02

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: like doing a semantic diff checking also between different representation formats. So we have a specification file and then a code and the test plans for it or stuff like that and then trying to figure out if the like intent to use like quote unquote um if the intent survives cuz one of the things I kind of encountered in my work in the past months is the the difference between like verification and validation. uh verification being like if we build something right and validation being if we built the

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: right thing and that's like a sort of um like the

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: specification file is basically the only one where humans fully still like write them. So like that's where the intent is put and then everything else especially if it's done with LLM then that's like a a projection or um like a a mapping into another another space

Claudiu Leoveanu-Condrei: There's there's also maybe this would be interesting for you.

Luiza Corpaci: with

Claudiu Leoveanu-Condrei: I obviously don't remember how it's called. Um there is like a protocol for business stuff.

01:33:26

Luiza Corpaci: your business.

Claudiu Leoveanu-Condrei: Mhm. It's something they they they use a lot. I mean, it should be here somewhere. I'm pretty sure. Come on. How many things I start recently, dude?

Luiza Corpaci: many.

Claudiu Leoveanu-Condrei: You kidding me? I'm sure I liked it because this is like a save. I'm using this to save stuff. I will find it anyways. I don't remember like how it's called but there is like a format a standard standardized format for mapping business logic. It's like a DSL if you know

Luiza Corpaci: Oh,

Claudiu Leoveanu-Condrei: DSLs.

Luiza Corpaci: okay.

Claudiu Leoveanu-Condrei: Yeah, maybe I didn't save it actually. Yeah, this is are quite old. Um I maybe I didn't save it. Well, I will send it to you to find it. I know how to find it.

Luiza Corpaci: Okay.

Claudiu Leoveanu-Condrei: This is going to take me some some

Luiza Corpaci: Thank Thank you. But does that feel like um I mean it it sort of does but like if for example if

01:35:04

Claudiu Leoveanu-Condrei: time.

Luiza Corpaci: I'm like the issue that I'm looking at for example is to look in between like a specific document that describes the implementation for something and then the

implementation and then figuring out

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: like um like a mapping based on which to like do yeah tests or sessions or like other other sort of artifacts. And then the semantic diff checker is basically the atomic thing that I'm kind of like trying to work on such that then I can go into seeing coverage or looking into the gaps or blind spots or stuff like that.

Claudiu Leoveanu-Condrei: You mean code coverage,

Luiza Corpaci: Um can you repeat the

Claudiu Leoveanu-Condrei: right?

Luiza Corpaci: question?

Claudiu Leoveanu-Condrei: Uh do you mean code coverage,

Luiza Corpaci: Yes.

Claudiu Leoveanu-Condrei: right?

Luiza Corpaci: Yes. Yes.

Claudiu Leoveanu-Condrei: Mhm.

Luiza Corpaci: Yes.

Claudiu Leoveanu-Condrei: Yeah, that's I don't think I understood how you do it, but uh yeah, I um I think the intermediate representation is something useful in this case. I mean, we're doing

Luiza Corpaci: Okay. Uh, I hadn't looked into the answer set programming so far,

01:36:10

Claudiu Leoveanu-Condrei: it

Luiza Corpaci: but I shall. Okay. Nice. Thank you.

Claudiu Leoveanu-Condrei: noise. Okay. So, yeah, sorry for the wrong link. Um,

Andrei Ungureanu: No worries. Okay.

Luiza Corpaci: We're good. Now we got to have two books.

Claudiu Leoveanu-Condrei: yeah, two books. Okay. Is there anything else that I I don't know I miss something or I yet another link that I that is broken.

Luiza Corpaci: I think so far so good.

Claudiu Leoveanu-Condrei: Cool. 10. I know

Andrei Ungureanu: Yeah. Hey, I'm good.

Claudiu Leoveanu-Condrei: Andre.

Andrei Ungureanu: I um was good even before. So now we have the correct

Claudiu Leoveanu-Condrei: Cool. Okay, guys.

Andrei Ungureanu: me

Claudiu Leoveanu-Condrei: Nice. Uh, see you then next week. Uh, I will send invites. Uh, I think I have all the emails. Um, yeah. and start start dropping the links and all of this into the into the GitHub so that we start having them.

Luiza Corpaci: Mhm.

Claudiu Leoveanu-Condrei: Now I even have them open which is good.

Andrei Ungureanu: Okay.

Claudiu Leoveanu-Condrei: Okay. And the notes these are the most important I think which I will be sharing. I think I can just export them from Google Docs Markdown. Cool. Well, thank you. It was nice to e meet you and uh keep in

Andrei Ungureanu: It's in here.

Luiza Corpaci: open.

Claudiu Leoveanu-Condrei: touch.

Andrei Ungureanu: Yeah,

Claudiu Leoveanu-Condrei: Looking forward to it.

Andrei Ungureanu: sure.

Claudiu Leoveanu-Condrei: Bye-bye.

Luiza Corpaci: I

Andrei Ungureanu: Byebye.

Transcription ended after 01:38:33

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